Morita Equivalence
Of Formal Poisson Manifolds



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Lie Algebroids and Poisson Manifolds

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On Morita Equivalence of Poisson Manifolds

Lie Algebroids and Poisson Geometry

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”Mathematics and mathematical science, as a garb of ideas [...] encompasses everything which [...] represents the life-world.”
— Edmund Husserli, *The Crisis of European Sciences and Transcendental Phenomenology* (1954)

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Introduction

A Poisson manifold $(M, \{\cdot,\cdot\})$ is, from the point of view of noncommutative geometry, a first-order approximation to a noncommutative space, as the bracket records the leading term of a would-be deformation of the commutative algebra $C^\infty(M)$ into a family of associative, generally noncommutative algebras. This heuristic is made precise by Kontsevich's formality theorem, which produces, for every Poisson structure π_0 on M, a star product $\star$ on $C^\infty(M)[[\lambda]]$ quantizing π_0, and, more generally, a bijection between formal Poisson structures $\pi = \sum_{k\geq 0} \lambda^k \pi_k$ up to a suitable notion of gauge equivalence and star products up to equivalence. Formal Poisson structures — Poisson bivectors valued in $\mathfrak{X}^2(M)[[\lambda]]$ satisfying $[\pi,\pi] = 0$ order by order in λ — therefore sit at the interface between classical Poisson geometry and deformation quantization, and their equivalence theory is the classical ghost of Morita theory for the associated star-product algebras.

On the classical side, two natural notions of equivalence compete for the role of "sameness" of Poisson manifolds. The first, *gauge equivalence*, is geometric and elementary: a closed 2-form $B \in \Omega^2_{\mathrm{cl}}(M)$ acts on Poisson structures transverse to its graph by the formula $\pi \mapsto e^B\pi$, a construction that can be understood entirely at the level of Dirac structures on $TM \oplus T^*M$. The second, *Morita equivalence*, is representation-theoretic in spirit: it declares (P_1,π_1) and (P_2,π_2) equivalent when they admit a full dual pair, a symplectic manifold (S,ω) realizing both as reduced spaces via a pair of surjective submersions with symplectically orthogonal, connected and simply connected fibers. Morita equivalence is the Poisson-geometric analogue of Morita equivalence of algebras, is designed to be invariant under the operations one expects to preserve "the same physics" (leaf spaces, isotropy Lie algebras, monodromy, Poisson cohomology), and reduces, in the symplectic case, to symplectomorphism. In general these two equivalence relations do not coincide, and understanding their relationship is a nontrivial problem already in the classical (non-formal) theory.

The purpose of this work is to give a complete, self-contained account of the theorem of Bursztyn, Ortiz and Waldmann [1] identifying Morita equivalence and gauge equivalence for formal Poisson structures vanishing at zeroth order, that is, deformations $\pi \in \lambda\,\mathfrak{X}^2(P)[[\lambda]]$ of the trivial Poisson structure $\pi_0 = 0$ on a manifold P. The main result, proved in Section 9 as Theorem 9.1, states that two such formal Poisson structures π and π' on P are Morita equivalent if and only if there is a diffeomorphism $\psi \in \mathrm{Diff}(P)$ carrying π into the orbit of $\psi_*\pi'$ under the gauge action of $H^2_{dR}(P)[[\lambda]]$. Formally, Morita equivalence collapses onto gauge equivalence modulo $\mathrm{Diff}(P)$, namely, the deformation parameter λ removes the obstructions that separate the two notions classically.

The proof assembles two independent pieces. The first, a *classification* statement (Theorem 8), shows that every self-equivalence bimodule of $(P,0)$ is realized, order by order in λ, by the cotangent Lie algebroid $T^*P \to P$ with a formal symplectic structure deforming the canonical one, and that the moduli of such structures is governed by $H^2_{dR}(P)[[\lambda]]$; this rests on the vanishing of a Chevalley–Eilenberg cohomology group controlling deformations of Poisson morphisms, which in turn is where the formality of λ becomes essential. The second, an *identification* statement (Theorem 9), computes the resulting classifying action explicitly, via a Dirac-geometric argument on T^*P, and shows that it agrees precisely with the B-field gauge action already introduced in Section 4. Composing the two yields the main theorem.

To make this argument precise, we develop in Section 1 the basic apparatus of Poisson geometry needed throughout: conventions, symplectic leaves and foliations, Poisson maps and submanifolds, Lie algebroids and groupoids, and fundamental objects like the cotangent Lie algebroid $(T^*M, [\cdot,\cdot]_\pi, \pi^\sharp)$. Section 2 introduces formal Poisson structures and their formal symplectic counterparts (with particular attention to the vanishing zeroth-order case that forms the core of this work), treats gauge transformations by closed 2-forms in detail — including their effect on symplectic leaves, Poisson cohomology, and symplectic groupoids — and establishes the framework of Morita equivalence of Poisson manifolds via symplectic bimodules alongside its topological, algebraic, and geometric invariants. Finally, Section 3 develops the deformation theory of Poisson morphisms on which the classification theorem depends, constructs the classifying action and its identification with gauge transformations, and combines these ingredients into the main result comparing Morita and gauge equivalences, closing with a discussion of future directions such as integral and Hochschild-cohomological refinements suggested by Kontsevich's quantization map.

Chapter 1

Foundations of Poisson Geometry

I.. Conventions and Basic Bivectors

This essay closely follows [1], a debt reflected, *inter alia*, in its expository architecture and choice of notation. Accordingly, throughout this work, we adopt the following conventions:

- The Schouten bracket on multivector fields has degree -1 and satisfies

$$,[X,Y] = -(-1)^{(|X|-1)(|Y|-1)}[Y,X],$$

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$$[X,Y] = -(-1)^{(|X|-1)(|Y|-1)}[Y,X],$$

  where $|X|$ denotes the degree of the multivector field X. A Poisson bivector π is characterized by the integrability condition

$$[\pi,\pi] = 0.$$

- For a symplectic manifold (S,ω), the Poisson bracket induced on functions is defined by

$$\{f,g\} = \omega(X_f, X_g),$$

  where X_f and X_g are the Hamiltonian vector fields associated to f and g, respectively. Consequently, for a symplectic leaf (S,ω_S) of a Poisson manifold (M,π), the leafwise symplectic form is given by

$$\omega_S(\pi^\sharp\alpha, \pi^\sharp\beta) = -\pi(\alpha,\beta), \qquad \alpha,\beta \in T^*M.$$

- On the generalized tangent bundle $\mathbb{T}M = TM \oplus T^*M$, the Courant-Dorfman bracket is defined by

$$[[X+\alpha, Y+\beta]] := [X,Y] + \mathscr{L}_X\beta - \iota_Y d\alpha,$$

  for vector fields $X, Y \in \mathfrak{X}(M)$ and 1-forms $\alpha,\beta \in \Omega^1(M)$.

Definition 1. *A **Poisson manifold** is a manifold M endowed with a Poisson bracket on the space $C^\infty(M)$ of smooth functions. This means a Lie bracket $\{\cdot,\cdot\} : C^\infty(M)\times C^\infty(M) \to C^\infty(M)$ satisfying Leibniz identity*

$$\{f, g\cdot h\} = g\cdot\{f,h\} + \{f,g\}\cdot h, \quad \forall f,g,h \in C^\infty(M).$$

Recall that $(C^\infty(M), \{\cdot,\cdot\})$ is a Lie algebra: the Poisson bracket is $\mathbb{R}$-bilinear and skew-symmetric, and satisfies the Jacobi identity,

$$\{f,\{g,h\}\} + \{h,\{f,g\}\} + \{g,\{h,f\}\} = 0, \quad \forall f,g,h \in C^\infty(M). \tag{1.1}$$

As stated in our conventions, the Poisson bivector π satisfies $[\pi,\pi] = 0$. For any $H \in C^\infty(M)$, the map $\{H,\cdot\}$ is a derivation of $C^\infty(M)$, and therefore determines a vector field $X_H \in \mathfrak{X}_{\mathrm{Ham}}(M)$ by the condition

$$\{H,f\} = \mathscr{L}_{X_H} f, \quad \forall f \in C^\infty(M). \tag{1.2}$$

The vector field X_H is called the Hamiltonian vector field associated to H. In physics, H is commonly taken to be the *Hamiltonian*, or the energy, of a given physical system. The set $\mathfrak{X}_{\mathrm{Ham}}(M)$ forms a Lie subalgebra.

Another interesting consequence of the Leibniz rule is that Poisson brackets are local, in the sense that they can be restricted to open sets $U \subset M$. So, in a local chart (U, x^i), the Poisson bracket takes the form

$$\{f,g\}|_U = \sum_{i,j=1}^{m} \pi^{ij} \frac{\partial f}{\partial x^i}\frac{\partial g}{\partial x^j},$$

for some smooth functions $\pi^{ij} \in C^\infty(U)$.

Definition 2. *A **bivector field** (or simply **bivector**) on a manifold M is a smooth section π of the bundle $\bigwedge^2 TM$, i.e., $\pi \in \mathfrak{X}^2(M) = \Gamma(\bigwedge^2 TM)$. In local coordinates (x^i), it is written as*

$$\pi = \sum_{i<j} \pi^{ij}(x) \frac{\partial}{\partial x^i} \wedge \frac{\partial}{\partial x^j},$$

with $\pi^{ij} \in C^\infty(M)$. A bivector induces a bundle map $\pi^\sharp : T^*M \to TM$ defined by $\langle \pi^\sharp(\alpha), \beta\rangle = \pi(\alpha,\beta)$.

Given a Poisson bracket $\{\cdot,\cdot\}$, the functions $\pi^{ij} = \{x^i, x^j\}$ define a bivector π. Conversely, a bivector π defines a bracket $\{f,g\} = \pi(df,dg)$. The Jacobi identity for the bracket is equivalent to $[\pi,\pi] = 0$, where $[\cdot,\cdot]$ is the Schouten bracket. Hence, a Poisson manifold can equivalently be defined as:

Definition 3. *A Poisson manifold is the pair (M,π), where M is a manifold and $\pi \in \mathfrak{X}^2(M)$, known as Poisson Structure, is a bivector field satisfying $[\pi,\pi] = 0$, where $[\cdot,\cdot] : \mathfrak{X}^{k+1}(M) \times \mathfrak{X}^{l+1} \to \mathfrak{X}^{k+l+1}$ is the Schouten braket.*

The following are some essential examples:

Example 1. *(**The Symplectic Case**) Consider the pair (M,ω), where M is a manifold and $\omega \in \Omega^2(M)$. If ω is nondegenerate ($\iota_X\omega = 0 \Rightarrow X = 0$) and closed ($d\omega = 0$), then $\pi = \omega^{-1}$ is a Poisson bivector. The bracket is the usual one of classical mechanics.*

Example 2. *(**Trivial example**) $\pi = 0$ gives the identically zero bracket.*

Example 3. *(**Linear Poisson structure**) Let $\mathfrak{g}$ be a Lie algebra. On $\mathfrak{g}^*$, the linear coordinates x^i satisfy $\{x^i,x^j\} = \sum_k c_k^{ij} x^k$, where c_k^{ij} are the structure constants. The symplectic foliation consists of the coadjoint orbits.*

II.. Symplectic leaves and the symplectic foliation

Let (M,π) be a Poisson manifold. Recall that the *orbits* of (M,π) are the equivalence classes of the relation $x \sim y$ if y can be reached from x by flowing along Hamiltonian vector fields. The fundamental structural theorem of Poisson geometry is the following.

Theorem 1 (Symplectic leaf theorem)**.** *Every orbit S of a Poisson manifold (M,π) carries a unique smooth structure making the inclusion $S \hookrightarrow M$ an immersion. The tangent spaces are*

$$T_xS = \operatorname{Im} \pi_x^\sharp, \qquad \forall\, x \in S,$$

and S carries a symplectic form ω_S defined by

$$\omega_S(\pi_x^\sharp\alpha,\ \pi_x^\sharp\beta) = -\pi_x(\alpha,\beta), \qquad \alpha,\beta \in T_x^*M. \tag{1.3}$$

Note that the sign convention in (1.3) matches our convention $\{f,g\} = \omega_S(X_f, X_g)$.

Definition 4. *A* symplectic leaf *of (M,π) is an orbit S together with the symplectic form ω_S from (1.3). The* symplectic foliation *of (M,π) is the collection*

$$\mathscr{F}_\pi := \{(S,\omega_S) : S \text{ is a symplectic leaf}\}.$$

Remark 1. *In general the leaves have varying dimension, so $\mathscr{F}_\pi$ is a* singular *foliation. A Poisson structure can be viewed as a partition of M into symplectic manifolds; conversely, a Poisson structure is completely determined by its symplectic foliation.*

Corollary 1. *The symplectic structure on the orbit S is the unique symplectic form ω_S for which the inclusion $(S,\omega_S^{-1}) \hookrightarrow (M,\pi)$ is a Poisson map.*

Proposition 1. *A connected immersed submanifold $i : N \to M$ with $T_xN = \operatorname{Im}\pi_x^\sharp$ for all $x \in N$ is an open subset of a single symplectic leaf.*

Regular Poisson structures.

Definition 5. (M,π) *is* regular *if* $\operatorname{rank}\pi$ *is constant on* M.

For a regular Poisson manifold, $\operatorname{Im}\pi^\sharp \subset TM$ is a smooth involutive subbundle, defining a regular foliation $\mathscr{F}_\pi$ with $T\mathscr{F}_\pi = \operatorname{Im}\pi^\sharp$. The leafwise symplectic forms assemble into a foliated 2-form $\omega_{\mathscr{F}_\pi} \in \Omega^2(\mathscr{F}_\pi)$ given by

$$\omega_{\mathscr{F}_\pi}(\pi^\sharp\alpha, \pi^\sharp\beta) = -\pi(\alpha,\beta).$$

Theorem 2. *Formulas* $T\mathscr{F}_\pi = \operatorname{Im}\pi^\sharp$ *and* $\omega_{\mathscr{F}_\pi}(\pi^\sharp\alpha,\pi^\sharp\beta) = -\pi(\alpha,\beta)$ *establish a bijection*

$$\{\text{regular Poisson structures } \pi \in \mathfrak{X}^2(M)\} \;\longleftrightarrow\; \{\text{symplectic foliations } (\mathscr{F},\omega_{\mathscr{F}}) \text{ on } M\}.$$

Examples. For a Lie algebra $\mathfrak{g}$ of a connected Lie group G, the symplectic leaves of the linear Poisson structure $(\mathfrak{g}^*,\pi_{\mathfrak{g}})$ are the *coadjoint orbits* $\mathcal{O}_\xi = G\cdot\xi$, each equipped with the *Kirillov–Kostant–Souriau* symplectic form

$$\omega_{\mathcal{O}_\xi}(\operatorname{ad}_v^*\xi,\ \operatorname{ad}_w^*\xi) := -\xi([v,w]), \qquad \xi\in\mathfrak{g}^*,\ v,w\in\mathfrak{g}.$$

III.. Poisson maps and Poisson submanifolds

Definition 6. *A smooth map* $\Phi : (M,\pi_M)\to(N,\pi_N)$ *between Poisson manifolds is a* Poisson map *if*

$$\{f\circ\Phi,\ g\circ\Phi\}_M = \{f,g\}_N\circ\Phi, \qquad \forall\, f,g\in C^\infty(N).$$

Equivalently, $d_x\Phi\circ(\pi_M)^\sharp_x = (\pi_N)^\sharp_{\Phi(x)}\circ(d_x\Phi)^*$ *for all* $x\in M$.

Remark 2. *Poisson maps send symplectic leaves to symplectic leaves. More precisely, if* $\Phi : (M,\pi_M)\to(N,\pi_N)$ *is a Poisson map and* $S\subset M$ *is a symplectic leaf, then* $\Phi(S)$ *is contained in a single symplectic leaf of* (N,π_N)*, and the restriction* $\Phi|_S : (S,\omega_S)\to(\Phi(S),\omega_{\Phi(S)})$ *is a symplectic map.*

Definition 7. *A* Poisson submanifold *of* (M,π) *is a Poisson manifold* (N,π_N) *together with an injective immersion* $i : N\to M$ *that is a Poisson map.*

The following characterises embedded Poisson submanifolds.

Proposition 2. *Let* (M,π) *be a Poisson manifold and* $N\hookrightarrow M$ *an immersed submanifold. There is at most one Poisson structure on* N *making it a Poisson submanifold, and it exists if and only if any of the following equivalent conditions hold:*

(i) $\operatorname{Im}\pi^\sharp_x \subset T_xN$ *for all* $x\in N$.

(ii) $(TN)^{\perp_\pi} := \pi^\sharp((TN)^\circ) = 0$.

(iii) *Every Hamiltonian vector field* X_H *is tangent to* N.

(iv) *The vanishing ideal* $I(N) := \{f\in C^\infty(M) : f|_N = 0\}$ *is a Lie ideal:* $\{f,g\}\in I(N)$ *whenever* $f\in I(N)$ *and* $g\in C^\infty(M)$ *(when* N *is embedded).*

In particular, a Poisson submanifold N *intersects each symplectic leaf* S *in an open subset of* S*, and the connected components of* $N\cap S$ *are the symplectic leaves of* (N,π_N).

Corollary 2. *If* $N_1,N_2\subset(M,\pi)$ *are Poisson submanifolds intersecting cleanly, then* $N_1\cap N_2$ *is also a Poisson submanifold.*

Complete Poisson submanifolds. An embedded Poisson submanifold $N\subset M$ is called *complete* if every Hamiltonian vector field X_H whose restriction to N is complete on N is itself complete on M. Complete Poisson submanifolds play a distinguished role in the global theory.

IV.. Lie Algebroids and Groupoids

Definition 8. *A* Lie algebroid *over a manifold* M *is a triple* $(A,[\cdot,\cdot]_A,\rho)$ *where*

- $A\to M$ *is a smooth vector bundle,*
- $[\cdot,\cdot]_A : \Gamma(A)\times\Gamma(A)\to\Gamma(A)$ *is a Lie bracket on sections,*
- $\rho : A\to TM$ *is a vector bundle map (the* anchor*),*

satisfying the Leibniz identity

$$[\alpha, f\beta]_A = f[\alpha,\beta]_A + \mathscr{L}_{\rho(\alpha)}(f)\,\beta, \qquad \forall\, \alpha,\beta \in \Gamma(A),\ f \in C^\infty(M).$$

Proposition 3. *For any Lie algebroid $(A, [\cdot,\cdot]_A, \rho)$, the anchor is a Lie algebra homomorphism:*

$$\rho([\alpha,\beta]_A) = [\rho(\alpha), \rho(\beta)], \qquad \forall\, \alpha,\beta \in \Gamma(A).$$

Proof. Consider the Jacobiator $J(\alpha,\beta,\gamma) := [\![[\![\alpha,\beta]_A,\gamma]_A + [\![[\![\beta,\gamma]_A,\alpha]_A + [\![[\![\gamma,\alpha]_A,\beta]_A$. Using only $\mathbb{R}$-bilinearity, skew-symmetry and the Leibniz identity, one computes

$$J(\alpha,\beta,f\gamma) - f\,J(\alpha,\beta,\gamma) = \mathscr{L}_{\rho([\alpha,\beta]_A)-[\rho(\alpha),\rho(\beta)]}(f)\,\gamma.$$

Since $[\cdot,\cdot]_A$ satisfies the Jacobi identity, $J = 0$, and the result follows. □

Associated to a Lie algebroid $(A, [\cdot,\cdot]_A, \rho)$ is the *Lie algebroid de Rham complex*: one defines A-differential forms as $\Omega^k(A) := \Gamma(\Lambda^k A^*)$, and the differential $d_A : \Omega^k(A) \to \Omega^{k+1}(A)$ by the Cartan formula

$$(d_A\omega)(\alpha_0,\dots,\alpha_k) = \sum_i (-1)^i \mathscr{L}_{\rho(\alpha_i)}(\omega(\alpha_0,\dots,\hat{\alpha}_i,\dots,\alpha_k)) + \sum_{i<j}(-1)^{i+j}\omega([\alpha_i,\alpha_j]_A,\alpha_0,\dots,\hat{\alpha}_i,\dots,\hat{\alpha}_j,\dots,\alpha_k).$$

One verifies $d_A^2 = 0$ and calls $H^\bullet(A)$ the resulting cohomology.

Example 4 (Tangent bundle)**.** *For any manifold M, the tangent bundle $A = TM$ is a Lie algebroid with the usual Lie bracket of vector fields and anchor $\rho = \mathrm{id}_{TM}$. The associated de Rham complex is the classical one.*

Example 5 (Lie algebras)**.** *A Lie algebra $\mathfrak{g}$ is a Lie algebroid over a point $M = \{*\}$: $A = \mathfrak{g}$, $[\cdot,\cdot]_A = [\cdot,\cdot]_{\mathfrak{g}}$, $\rho = 0$.*

Example 6 (Action Lie algebroid)**.** *Let $\mathfrak{g}$ be a Lie algebra acting on M by a Lie algebra homomorphism $a : \mathfrak{g} \to \mathfrak{X}(M)$. The* action Lie algebroid *is $A = M \times \mathfrak{g} \to M$ (trivial bundle with fiber $\mathfrak{g}$) with anchor*

$$\rho : M \times \mathfrak{g} \to TM, \quad (x,v) \mapsto a(v)_x,$$

and Lie bracket on $\Gamma(A) \cong C^\infty(M;\mathfrak{g})$ given by

$$[f,g](x) = [f(x),g(x)]_{\mathfrak{g}} + \mathscr{L}_{a(f(x))}g(x) - \mathscr{L}_{a(g(x))}f(x).$$

Example 7 (Foliations)**.** *An involutive distribution $D \subset TM$ is a Lie algebroid with bundle $A = D$, anchor the inclusion $\rho : D \hookrightarrow TM$, and bracket the restriction of the Lie bracket of vector fields.*

Example 8 (Cotangent algebroid of a Poisson manifold)**.** *For a Poisson manifold (M,π), the cotangent bundle $A = T^*M$ is a Lie algebroid with*

$$\rho = \pi^\sharp : T^*M \to TM, \qquad [\alpha,\beta]_\pi = \mathscr{L}_{\pi^\sharp\alpha}\beta - \mathscr{L}_{\pi^\sharp\beta}\alpha - d\pi(\alpha,\beta).$$

This is, without any suprise, known as the cotangent Lie algebroid. The orbits of this algebroid (i.e., the connected components of $\bigcup_x \operatorname{Im}\rho_x$) are exactly the symplectic leaves of (M,π).

Example 9 (Coisotropic submanifolds)**.** *A coisotropic submanifold $C \subset (M,\pi)$ satisfies $\pi^\sharp((TC)^\circ) \subset TC$. The conormal bundle $\nu^*(C) := (TC)^\circ \subset T^*M|_C$ inherits a Lie algebroid structure with anchor $\pi^\sharp|_{\nu^*(C)} : \nu^*(C) \to TC$ and bracket induced by $[\cdot,\cdot]_\pi$.*

Definition 9. *A* groupoid *consists of a set M of* objects *and a set G of* arrows*, together with:*

(i) *source and target maps $s,t : G \to M$,*

(ii) *a partially defined multiplication $m : G^{(2)} \to G, (g,h) \mapsto gh$, on composable pairs $G^{(2)} := \{(g,h) \in G\times G : s(g) = t(h)\}$, satisfying $s(gh) = s(h)$, $t(gh) = t(g)$, and associativity,*

(iii) *a unit map $u : M \to G$, $x \mapsto 1_x$, with $s(1_x) = t(1_x) = x$ and $g \cdot 1_{s(g)} = 1_{t(g)} \cdot g = g$,*

(iv) *an inverse map $\iota : G \to G$, $g \mapsto g^{-1}$, with $s(g^{-1}) = t(g)$, $t(g^{-1}) = s(g)$, and $g^{-1}g = 1_{s(g)}$, $gg^{-1} = 1_{t(g)}$.*

One writes $g : x \to y$ to indicate $s(g) = x$, $t(g) = y$, and denotes the groupoid $G \rightrightarrows M$.

Definition 10. *A* Lie groupoid *is a groupoid $G \rightrightarrows M$ where G and M are smooth manifolds, s and t are submersions, and the maps m, u, ι are smooth.*

Remark 3. *Since s and t are submersions, $G^{(2)} \subset G \times G$ is a smooth submanifold, so the smoothness of $m : G^{(2)} \to G$ is well-posed. One can show that m is a submersion, u is a closed embedding, and ι is a diffeomorphism.*

For a Lie groupoid $G \rightrightarrows M$ and $x \in M$, one defines:

- s-fibers and t-fibers: $s^{-1}(x)$, $t^{-1}(x)$ (submanifolds of G),
- the *isotropy Lie group* at x: $G_x := s^{-1}(x) \cap t^{-1}(x)$,
- the *orbit* of x: $\mathcal{O}_x := t(s^{-1}(x)) \subset M$ (immersed submanifold),
- *right* and *left translations*: $R_g : s^{-1}(t(g)) \to s^{-1}(s(g))$, $h \mapsto hg$, and $L_g : t^{-1}(s(g)) \to t^{-1}(t(g))$, $h \mapsto gh$.

The orbits partition M into the *orbit foliation* of $G \rightrightarrows M$.

Definition 11. *A* Lie groupoid morphism *from $G \rightrightarrows M$ to $H \rightrightarrows N$ is a pair of smooth maps $\Phi : G \to H$ and $\varphi : M \to N$ commuting with all structure maps and satisfying $\Phi(gh) = \Phi(g)\Phi(h)$.*

Example 10 (Pair groupoid)**.** *For a manifold M, the* pair groupoid *is $M \times M \rightrightarrows M$ with $s(y,x) = x$, $t(y,x) = y$, multiplication $(z,y)\cdot(y,x) = (z,x)$, units $1_x = (x,x)$, and inverses $(y,x)^{-1} = (x,y)$. Every two points are connected by a unique arrow; all isotropy groups are trivial and there is a single orbit.*

Example 11 (Action groupoid)**.** *Let a Lie group G act on M from the left. The* action groupoid *is $G \times M \rightrightarrows M$ with*

$$s(g,x) = x, \quad t(g,x) = g\cdot x, \quad (h, g\cdot x)\cdot(g,x) = (hg,x), \quad 1_x = (e,x), \quad (g,x)^{-1} = (g^{-1}, g\cdot x).$$

The orbits of this groupoid are the G-orbits in M, and the isotropy group at x is $G_x = \{g \in G : g\cdot x = x\}$.

Example 12 (Fundamental groupoid)**.** *For a manifold M, the* fundamental groupoid $\Pi(M) \rightrightarrows M$ *has as arrows the homotopy classes (relative endpoints) of paths in M:*

$$s([\gamma]) = \gamma(0), \quad t([\gamma]) = \gamma(1), \quad [\gamma]\cdot[\sigma] = [\gamma * \sigma],$$

where $\gamma * \sigma$ denotes concatenation (with $\gamma(0) = \sigma(1)$). The isotropy group at x is the fundamental group $\pi_1(M,x)$, and there is a single orbit if M is connected.

Example 13 (Submersion groupoid and holonomy groupoid)**.** *Given a submersion $\mu : M \to N$, the* submersion groupoid *$M \times_\mu M \rightrightarrows M$ consists of pairs (y,x) with $\mu(y) = \mu(x)$, with the same structure as the pair groupoid restricted to fibers of μ.*

Given a foliation $\mathcal{F}$ on M, the homotopy groupoid *$\Pi(M,\mathcal{F}) \rightrightarrows M$ has as arrows leafwise homotopy classes of paths lying in leaves, and the* holonomy groupoid *$\mathrm{Hol}(M,\mathcal{F}) \rightrightarrows M$ is its quotient by the holonomy equivalence relation. There is a surjective groupoid morphism $\Pi(M,\mathcal{F}) \twoheadrightarrow \mathrm{Hol}(M,\mathcal{F})$.*

Analogously to how a Lie group determines a Lie algebra, a Lie groupoid $G \rightrightarrows M$ determines a Lie algebroid.

Definition 12. *A vector field $X \in \mathfrak{X}(G)$ is* left-invariant *if*

(i) *X is tangent to the t-fibers of G,*

(ii) *$d_h L_g(X_h) = X_{gh}$ for all composable $(g,h) \in G^{(2)}$.*

Let $\mathfrak{X}_{\mathrm{inv}}(G)$ denote the space of left-invariant vector fields.

Left-invariant vector fields are in bijection with sections of the restricted bundle $u^*\ker(dt) \to M$.

Definition 13. *The* Lie algebroid of *$G \rightrightarrows M$ is the vector bundle*

$$A := u^*\ker(dt) \to M,$$

with anchor $\rho_x := d_{1_x}s : A_x = (\ker dt)|_{1_x} \to T_xM$ and Lie bracket

$$[\alpha_1,\alpha_2]_A \in \Gamma(A) \quad \text{defined by} \quad \overleftarrow{[\alpha_1,\alpha_2]_A} = [\overleftarrow{\alpha_1}, \overleftarrow{\alpha_2}],$$

where $\overleftarrow{\alpha}$ denotes the left-invariant vector field extending $\alpha \in \Gamma(A)$.

Theorem 3. *$(A, [\cdot,\cdot]_A, \rho)$ is a Lie algebroid. Moreover, a Lie groupoid morphism $\Phi : G \to H$ induces a Lie algebroid morphism $\Phi_* = d\Phi|_{u^*\ker(dt)} : \mathrm{Lie}(G) \to \mathrm{Lie}(H)$.*

Example 14. *The Lie algebroid of the pair groupoid $M \times M \rightrightarrows M$ is the tangent bundle TM. The Lie algebroid of the action groupoid $G \times M \rightrightarrows M$ is the action Lie algebroid $M \times \mathfrak{g}$. The Lie algebroid of $\Pi(M) \rightrightarrows M$ is again TM.*

V.. Morphisms, Integration, and Cohomology

Definition 14. *A* Lie algebroid morphism *from $(A, [\cdot,\cdot]_A, \rho_A)$ over M to $(B, [\cdot,\cdot]_B, \rho_B)$ over N is a vector bundle map $\phi : A \to B$ covering a smooth map $\varphi : M \to N$ such that*

$$\phi^* \circ d_B = d_A \circ \phi^*,$$

i.e., ϕ commutes with the Lie algebroid differentials. Equivalently, for all $\alpha, \beta \in \Gamma(A)$,

$$\phi([\alpha,\beta]_A) = [\phi(\alpha), \phi(\beta)]_B, \qquad \rho_B \circ \phi = d\varphi \circ \rho_A.$$

Definition 15. *A Lie algebroid $A \to M$ is* integrable *if there exists a Lie groupoid $G \rightrightarrows M$ and a Lie algebroid isomorphism $A \cong \mathrm{Lie}(G)$. Such G is called an* integration *of A.*

Remark 4. *Unlike the classical Lie third theorem for Lie algebras, not every Lie algebroid is integrable. The precise obstruction is cohomological and is described in terms of the* monodromy groups *of A (cf. Chapter 14 of [7]). A Lie algebroid is integrable if and only if its monodromy groups are locally uniformly discrete.*

The Lie philosophy for groupoids and algebroids can be summarized as follows:

- The functor $G \mapsto \mathrm{Lie}(G)$ from simply connected Lie groupoids to integrable Lie algebroids is essentially surjective.
- If $G \rightrightarrows M$ is simply connected (i.e., s-simply connected: all s-fibers are simply connected), then it is the unique integration of $\mathrm{Lie}(G)$ with this property.

The obstruction to integration motivates a general construction of the *Weinstein groupoid* (the monodromy or source-simply-connected groupoid) via A-paths.

Definition 16. *Let $A \to M$ be a Lie algebroid with anchor ρ. An* A-path *is a smooth path $a : [0,1] \to A$ such that*

$$\rho(a(t)) = \frac{d\gamma}{dt}(t), \qquad \text{where } \gamma = \pi_A \circ a : [0,1] \to M.$$

The path γ is called the base path *of a.*

Remark 5. *For $A = TM$ (tangent bundle), every TM-path is just an ordinary path in M (with $a(t) = \dot\gamma(t)$). For the cotangent algebroid $A = T^*M$ of a Poisson manifold, T^*M-paths are called* cotangent paths *and satisfy $\pi^\sharp(a(t)) = \dot\gamma(t)$.*

Definition 17. *Two A-paths a_0 and a_1 with the same endpoints $\gamma_i(0) = x$ and $\gamma_i(1) = y$ are* A-homotopic *if there exists a smooth family a_ε, $\varepsilon \in [0,1]$, of A-paths connecting them, together with a smooth family $b_\varepsilon : [0,1] \to A$ of A-paths satisfying appropriate compatibility conditions. One writes $a_0 \sim_A a_1$.*

Definition 18. *The* Weinstein groupoid *of a Lie algebroid $A \to M$ is*

$$\Pi(A) := \frac{\{A\text{-paths}\}}{\sim_A} \rightrightarrows M,$$

with source $s([a]) = \gamma(0)$, target $t([a]) = \gamma(1)$, and multiplication induced by concatenation of A-paths.

Theorem 4 (Crainic–Fernandes)**.** *The Weinstein groupoid $\Pi(A) \rightrightarrows M$ is a topological groupoid. It is a (possibly non-Hausdorff) Lie groupoid if and only if A is integrable, in which case it is the unique source-simply-connected integration of A up to isomorphism.*

Example 15 (Poisson homotopy groupoid)**.** *For a Poisson manifold (M,π), the Weinstein groupoid of the cotangent algebroid T^*M is the* Poisson homotopy groupoid

$$\Pi(M,\pi) = \frac{\{\text{cotangent paths}\}}{\text{cotangent path-homotopy}} \rightrightarrows M.$$

Its orbits are the symplectic leaves of (M,π), and its isotropy groups are the Poisson homotopy groups. The integrability of (M,π) as a Poisson manifold (i.e., the existence of an integrating symplectic groupoid) is equivalent to $\Pi(M,\pi)$ being a smooth Lie groupoid.

Let (M,π) be a Poisson manifold. Recall that the cotangent Lie algebroid T^*M has de Rham complex $(\Omega^\bullet(T^*M), d_{T^*M})$. Under the identification $\Omega^k(T^*M) = \Gamma(\Lambda^k TM) = \mathfrak{X}^k(M)$, this differential becomes:

Definition 19. *The* Poisson differential *is the map $d_\pi : \mathfrak{X}^k(M) \to \mathfrak{X}^{k+1}(M)$ defined by*

$$d_\pi \vartheta(\alpha_0,\dots,\alpha_k) = \sum_{i=0}^{k} (-1)^i \mathscr{L}_{\pi^\sharp(\alpha_i)}(\vartheta(\alpha_0,\dots,\hat{\alpha}_i,\dots,\alpha_k)) + \sum_{0\le i<j\le k} (-1)^{i+j} \vartheta([\alpha_i,\alpha_j]_\pi, \alpha_0,\dots,\hat{\alpha}_i,\dots,\hat{\alpha}_j,\dots,\alpha_k). \tag{1.4}$$

Proposition 4. *The Poisson differential satisfies $d_\pi \vartheta = [\pi, \vartheta]$, where $[\cdot,\cdot]$ denotes the Schouten–Nijenhuis bracket. In particular, $d_\pi^2 = 0$.*

Proof. By Exercise 9.12 of the reference, the formula (1.4) coincides with $[\pi, \vartheta]$. Then $d_\pi^2 \vartheta = [\pi, [\pi, \vartheta]] = \frac{1}{2}[[\pi,\pi], \vartheta] = 0$, since $[\pi, \pi] = 0$ is the Poisson condition. □

Definition 20. *The* Poisson cohomology *of (M, π) is the cohomology of the complex $(\mathfrak{X}^\bullet(M), d_\pi)$:*

$$H_\pi^k(M) := \frac{\ker(d_\pi : \mathfrak{X}^k(M) \to \mathfrak{X}^{k+1}(M))}{\mathrm{Im}(d_\pi : \mathfrak{X}^{k-1}(M) \to \mathfrak{X}^k(M))}.$$

The graded space $H_\pi^\bullet(M) = \bigoplus_k H_\pi^k(M)$ inherits additional algebraic structure.

Proposition 5. *The wedge product and the Schouten bracket on multivector fields descend to Poisson cohomology:*

$$H_\pi^p(M) \times H_\pi^q(M) \to H_\pi^{p+q}(M), \quad [\vartheta_1] \wedge [\vartheta_2] := [\vartheta_1 \wedge \vartheta_2],$$

$$H_\pi^{k+1}(M) \times H_\pi^{l+1}(M) \to H_\pi^{k+l+1}(M), \quad \langle [\vartheta_1], [\vartheta_2] \rangle := [[\vartheta_1, \vartheta_2]].$$

In particular, $H_\pi^\bullet(M)$ is a graded commutative algebra and a graded Lie algebra (with degree shift $p \mapsto k+1$), with the two structures related by the graded Leibniz identity.

In what follows, we study some low degree results.

Degree 0. The differential $d_\pi : C^\infty(M) \to \mathfrak{X}(M)$ sends $f \mapsto [\pi, f] = -X_f$. Therefore,

$$H_\pi^0(M) = \{f \in C^\infty(M) : \{f, g\} = 0 \ \forall g \in C^\infty(M)\} =: \mathrm{Cas}(M, \pi),$$

the space of *Casimir functions.* This is a ring under pointwise multiplication, typically infinite dimensional.

Degree 1. The differential $d_\pi : \mathfrak{X}(M) \to \mathfrak{X}^2(M)$ sends $X \mapsto [\pi, X] = -\mathscr{L}_X \pi$. Hence $\ker d_\pi$ consists of *Poisson vector fields* (infinitesimal automorphisms), and $\mathrm{Im}\, d_\pi$ consists of *Hamiltonian vector fields*:

$$H_\pi^1(M) \cong \frac{\mathfrak{X}(M, \pi)}{X_{\mathrm{Ham}}(M, \pi)}.$$

This is the Lie algebra of *outer Poisson derivations* of $(C^\infty(M), \{\cdot,\cdot\})$.

Degree 2. In degree 2, the differential is $d_\pi : \mathfrak{X}^2(M) \to \mathfrak{X}^3(M)$, $\vartheta \mapsto [\pi, \vartheta]$. Thus

$$H_\pi^2(M) \cong \frac{\{\vartheta \in \mathfrak{X}^2(M) : [\pi, \vartheta] = 0\}}{\{\mathscr{L}_X \pi : X \in \mathfrak{X}(M)\}}.$$

The Poisson bivector itself defines a class $[\pi] \in H_\pi^2(M)$, called the *fundamental class.* The Poisson manifold is *exact* if $[\pi] = 0$, i.e., if $\pi = \mathscr{L}_X \pi$ for some vector field X.

Suppose (M, π) is orientable and let $\mu \in \Omega^{\mathrm{top}}(M)$ be a volume form. For any $f \in C^\infty(M)$, the Lie derivative $\mathscr{L}_{X_f} \mu$ is a top-degree form, so

$$\mathscr{L}_{X_f} \mu = X_\mu(f)\, \mu$$

for a uniquely determined function $X_\mu(f)$.

Lemma V..1. *The assignment $f \mapsto X_\mu(f)$ is a derivation of $C^\infty(M)$, hence defines a vector field $X_\mu \in \mathfrak{X}(M)$ called the* modular vector field *of (M, π) relative to μ. Moreover:*

(i) *X_μ is a Poisson vector field: $[X_\mu, \pi] = 0$.*

(ii) *If $\mu' = e^g \mu$ is another volume form, then $X_{\mu'} = X_\mu - X_g$, a change by a Hamiltonian vector field.*

Definition 21. *The* modular class *of an orientable Poisson manifold (M, π) is*

$$\mathrm{mod}(M, \pi) := [X_\mu] \in H_\pi^1(M).$$

This class is independent of the choice of volume form μ. The Poisson manifold is called unimodular *if* $\mathrm{mod}(M, \pi) = 0$.

Corollary 3. *(M, π) is unimodular if and only if it admits a volume form invariant under all Hamiltonian diffeomorphisms.*

Example 16 (Linear Poisson structures)**.** *A Lie algebra* $\mathfrak{g}$ *is called* unimodular *if* $\mathrm{tr}(\mathrm{ad}_v) = 0$ *for all* $v \in \mathfrak{g}$*. The linear Poisson structure* $(\mathfrak{g}^*, \pi_{\mathfrak{g}})$ *is unimodular as a Poisson manifold if and only if* $\mathfrak{g}$ *is unimodular as a Lie algebra.*

Example 17 (Log-symplectic structures)**.** *For an orientable log-symplectic Poisson manifold* (M^{2n}, π) *with nonempty singular locus* Z*, fix a volume form* μ*. The modular vector field* X_μ *extends smoothly to* Z *and is nontrivial there, so* (M, π) *is not unimodular (unless* $Z = \emptyset$*).*

Definition 22. *A* deformation *of a Poisson structure* π *on* M *is a smooth family* $\{\pi_t\}_{t\in I}$ *of Poisson bivectors on* M*, parameterised by an interval* $I \ni 0$*, with* $\pi_0 = \pi$*. The* variation *at* $t = 0$ *is*

$$\vartheta := \left.\frac{d}{dt}\right|_{t=0} \pi_t \in \mathfrak{X}^2(M).$$

Differentiating the Maurer–Cartan equation $[\pi_t, \pi_t] = 0$ at $t = 0$ yields $[\pi, \vartheta] = 0$, i.e., $d_\pi \vartheta = 0$. Thus every variation of a deformation defines a *cocycle* $\vartheta \in \ker d_\pi \subset \mathfrak{X}^2(M)$.

Definition 23. *Two deformations* $\{\pi_t\}$ *and* $\{\pi'_t\}$ *of* π *are* equivalent *if there exists a smooth family of diffeomorphisms* $\{\varphi_t\}_{t\in I}$ *with* $\varphi_0 = \mathrm{id}$ *and* $\pi'_t = (\varphi_t)_* \pi_t$*. The starting velocity* $X \in \mathfrak{X}(M)$ *of* $\{\varphi_t\}$ *satisfies* $\vartheta' - \vartheta = \mathscr{L}_X \pi = -d_\pi X$*.*

Proposition 6. *The cohomology class* $[\vartheta] \in H^2_\pi(M)$ *depends only on the equivalence class of the deformation. In particular, the second Poisson cohomology* $H^2_\pi(M)$ *classifies infinitesimal deformations of* π *modulo equivalence.*

Remark 6. *Not every class in* $H^2_\pi(M)$ *integrates to a genuine deformation. The graded Lie bracket on* $H^\bullet_\pi(M)$ *provides an obstruction: if* $c \in H^2_\pi(M)$ *is the class of a deformation, then* $[c, c] = 0 \in H^3_\pi(M)$*. More generally, the problem of formal integration is governed by an* L_∞*-algebra structure on* $(\mathfrak{X}^\bullet(M)[1], [\cdot,\cdot])$*.*

Example 18 (Fundamental class)**.** *The family* $\pi_t := e^t \pi = (1 + t)\pi + O(t^2)$ *(or more precisely* $\pi_t := e^t \pi$ *for* t *near* 0*) is a deformation of* π *with variation* $\vartheta = \pi$*, giving the fundamental class* $[\pi] \in H^2_\pi(M)$*.*

Example 19 (Symplectic case)**.** *For a symplectic manifold* (M, ω)*, identifying* $\pi = \omega^{-1}$*, we have the isomorphism* $H^k_\pi(M) \cong H^{2n-k}_{\mathrm{dR}}(M)$ *(where* $\dim M = 2n$*) through the map* $\vartheta \mapsto \iota_\vartheta(\omega^n/n!)$*. In particular,* $H^2_\pi(M) \cong H^{2n-2}_{\mathrm{dR}}(M)$*, which is the cohomological home of symplectic deformations.*

The generalized tangent bundle of a manifold M is $\mathbb{T}M := TM \oplus T^*M$, equipped with the symmetric pairing

$$\langle X \oplus \alpha, Y \oplus \beta \rangle := \alpha(Y) + \beta(X)$$

and the Courant-Dorfman bracket

$$[[X \oplus \alpha, Y \oplus \beta]] := [X, Y] \oplus (\mathscr{L}_X \beta - \iota_Y d\alpha).$$

A **Dirac structure** is a subbundle $L \subset \mathbb{T}M$ which is: 1. **Maximal isotropic**: $L = L^\perp$ with respect to $\langle \cdot, \cdot \rangle$ 2. **Involutive**: $[[L, L]] \subset L$

Every Poisson structure π defines a Dirac structure via its graph:

$$L_\pi := \{\pi^\sharp(\alpha) \oplus \alpha \,:\, \alpha \in T^*M\}.$$

Conversely, Dirac structures with $L \cap TM = 0$ correspond to Poisson bivectors.

Chapter 2

Some Foundational Results

I.. Preliminaries

We begin this section with a recapitulation and further elaboration of several concepts already introduced. A Poisson map $\varphi : (P_1, \pi_1) \to (P_2, \pi_2)$ is **complete** if, whatever X_f is a complete vector field for $f \in C^\infty(P_2)$ — that is, the flow $\Phi_t^f : P_2 \to P_2$ generated by f exists for all $t \in \mathbb{R}$ —, then so is $X_{\varphi^* f}$. A map $\varphi : (P_1, \pi) \to (P_2, -\pi_2)$ is **anti-Poisson** if $\varphi : (P_1, \pi) \to (P_2, \pi_2)$ is Poisson. The formal space of power series in λ (over $\mathbb{K}[[\lambda]]$),

$$\sum_{k=1}^{\infty} \lambda^k f_k,$$

with $f_k \in C^\infty(P)$, is denoted by $C^\infty(P)[[\lambda]]$. The same applies if we replace $C^\infty(P)$ with $\mathfrak{X}^\bullet(P)$ or with $\Omega^\bullet(P)$. The formal spaces $C^\infty(P)[[\lambda]]$, $\Omega^\bullet(P)[[\lambda]]$, and $\mathfrak{X}^\bullet(P)[[\lambda]]$ each carry a natural algebra structure over $\mathbb{K}[[\lambda]]$. To illustrate this construction, consider the case of $\Omega^\bullet(P)[[\lambda]]$. Let $\omega_\lambda, \omega'_\lambda \in \Omega^\bullet(P)[[\lambda]]$ be given by the formal series

$$\omega_\lambda = \sum_{k=0}^{\infty} \lambda^k \omega_k, \qquad \omega'_\lambda = \sum_{\ell=0}^{\infty} \lambda^\ell \omega'_\ell,$$

where $\omega_k, \omega'_\ell \in \Omega^\bullet(P)$. Since $\Omega^\bullet(P)$ is endowed with the wedge product, this product extends naturally by $\mathbb{K}[[\lambda]]$-bilinearity to $\Omega^\bullet(P)[[\lambda]]$ via

$$\omega_\lambda \wedge \omega'_\lambda = \left(\sum_{k=0}^{\infty} \lambda^k \omega_k\right) \wedge \left(\sum_{\ell=0}^{\infty} \lambda^\ell \omega'_\ell\right) = \sum_{n=0}^{\infty} \lambda^n \sum_{k=0}^{n} \omega_k \wedge \omega'_{n-k}.$$

Since the wedge product on $\Omega^\bullet(P)$ is associative and graded-commutative, $\Omega^\bullet(P)[[\lambda]]$ inherits the structure of a graded $\mathbb{K}[[\lambda]]$-algebra. The constructions for $C^\infty(P)[[\lambda]]$ and $\mathfrak{X}^\bullet(P)[[\lambda]]$ are entirely analogous, using respectively the pointwise product and the exterior product of multivector fields. Moreover, the de Rham differential $d : \Omega^\bullet(P) \to \Omega^{\bullet+1}(P)$ extends λ-linearly to $\Omega^\bullet(P)[[\lambda]]$: given $\omega_\lambda \in \Omega^\bullet(P)[[\lambda]]$, one sets

$$d\,\omega_\lambda := \sum_{k=0}^{\infty} \lambda^k\, d\omega_k.$$

Likewise, the Schouten bracket extends λ-linearly to $\mathfrak{X}^\bullet(P)[[\lambda]]$: given

$$X = \sum_{k=0}^{\infty} \lambda^k X_k, \qquad Y = \sum_{\ell=0}^{\infty} \lambda^\ell Y_\ell \in \mathfrak{X}^\bullet(P)[[\lambda]],$$

one sets

$$[X, Y] := \sum_{k,\ell=0}^{\infty} \lambda^{k+\ell} [X_k, Y_\ell].$$

For a formal vector field $X \in \lambda\mathfrak{X}(P)[[\lambda]]$, one defines a **formal diffeomorphism** on P as the formal power series

$$\exp(\mathscr{L}_X) := \mathrm{id} + \sum_{i=1}^{\infty} \frac{\mathscr{L}_X^i}{i!},$$

where $\mathscr{L}_X = \sum_{i=1}^{\infty} \lambda^i \mathscr{L}_{X_i}$ is the Lie derivative along X. Such formal diffeomorphisms form a group. Indeed, the Lie derivatives $\mathscr{L}_X$ associated with formal vector fields $X \in \mathfrak{X}(P)[[\lambda]]$ constitute a Lie algebra under the commutator bracket, since

$$[\mathscr{L}_X, \mathscr{L}_Y] = \mathscr{L}_{[X,Y]}.$$

Consequently, the Baker–Campbell–Hausdorff formula implies that the composition of two formal diffeomorphisms is again of the same form:

$$\exp(\mathscr{L}_X)\exp(\mathscr{L}_Y) = \exp(\mathscr{L}_Z),$$

where Z is given by the Baker–Campbell–Hausdorff series in X and Y. The identity element is $\mathrm{id} = \exp(0)$, whilst the inverse of $\exp(\mathscr{L}_X)$ is $\exp(-\mathscr{L}_X)$. Hence, the set of formal diffeomorphisms defines a group acting on $C^\infty(P)[[\lambda]]$, $\Omega^\bullet(P)[[\lambda]]$, and $\mathfrak{X}^\bullet(P)[[\lambda]]$, preserving their algebraic structures and, in the latter two cases, the de Rham differential and the Schouten bracket, respectively.

Let (P, π_P) and (S, π_S) be Poisson manifolds, and consider the corresponding $\mathbb{K}[[\lambda]]$-algebras $C^\infty(P)[[\lambda]]$ and $C^\infty(S)[[\lambda]]$, each endowed with the natural commutative product induced by pointwise multiplication. In this setting, any morphism of commutative $\mathbb{K}[[\lambda]]$-algebras $\Phi : C^\infty(P)[[\lambda]] \longrightarrow C^\infty(S)[[\lambda]]$ may be expressed as a formal power series of linear maps

$$\Phi = \sum_{j=0}^{\infty} \lambda^j \phi_j,$$

where each $\phi_j : C^\infty(P) \to C^\infty(S)$ is $\mathbb{K}$-linear. Furthermore, the zeroth-order term

$$\phi_0 : C^\infty(P) \longrightarrow C^\infty(S)$$

is itself a morphism of commutative algebras. By the standard correspondence between smooth maps and algebra morphisms of smooth functions, there exists a smooth map

$$J : S \longrightarrow P$$

such that

$$\phi_0 = J^*.$$

In particular, when $S = P$ and $\phi_0 = \mathrm{id}$, the morphism Φ is a formal diffeomorphism. In this case, it may be written in the form

$$\Phi = \exp(\mathscr{L}_X),$$

for some formal vector field $X \in \lambda\mathfrak{X}(P)[[\lambda]]$. The justification lies on the following lemma:

Lemma I..1. *Let*

$$\Phi = J^* + \sum_{k=1}^{\infty} : C^\infty(P)[[\lambda]] \to C^\infty(S)[[\lambda]]$$

be a morphism of commutative rings, with $J : S \to P$ a surjective submersion. Then, there exists a formal vector field $Z \in \lambda\mathfrak{X}(S)[[\lambda]]$ such that $\Phi = \exp(\mathscr{L}_Z)J^*$.

This forms the base where we will construct the formal deformation of the Poisson morphism and, consequently, of the Morita equivalence of Poisson structures in the following sections.

1.. Formal Poisson Structures

Definition 24. *Let P be a manifold. A **formal Poisson structure** on P is a formal series*

$$\pi = \sum_{i=0}^{\infty} \lambda^i \pi_i \in \mathfrak{X}^2(P)[[\lambda]]$$

such that it satisfies the integrability condition

$$[\pi, \pi] = \left[\sum_{i=0}^{\infty} \lambda^i \pi_i, \sum_{j=0}^{\infty} \lambda^j \pi_j\right] = \sum_{i,j=0}^{\infty} \lambda^{i+j}[\pi_i, \pi_j],$$

with $[\cdot,\cdot]$ the λ-bilinearly extended Schouten bracket.

In the ordinary setting, a Poisson structure may be regarded as the Poisson bracket on the ring $C^\infty(P)$ of smooth functions. Analogously, a formal Poisson structure may be regarded as a formal Poisson bracket on the formal Poisson manifold, given by

$$\{f, g\}_\pi = \sum_{k=0}^{\infty} \lambda^k \sum_{i+j=k} \pi_i(df_j, dg_{k-j}),$$

where $f = \sum_{i=0}^{\infty} \lambda^i f_i,\ g = \sum_{j=0}^{\infty} \lambda^j g_j \in C^\infty(P)[[\lambda]]$ are formal smooth functions. The formal bracket $\{\cdot,\cdot\}_\pi$ extends the classical Poisson bracket induced by π_0. In particular, its zeroth-order term is given by

$$\{f, g\}_\pi\big|_{\lambda=0} = \pi_0(df, dg),$$

so that π_0 defines an ordinary Poisson structure on P.

Two formal Poisson structures $\pi^{(1)}$ and $\pi^{(2)}$ are said to be equivalent if there exists a formal vector field $X \in \lambda\mathfrak{X}(P)[[\lambda]]$ such that

$$\pi^{(2)} = \exp(\mathscr{L}_X)\,\pi^{(1)}.$$

Equivalently, this is to say that the map

$$\exp(\mathscr{L}_X) : \left(C^\infty(P)[[\lambda]], \{\cdot,\cdot\}^{(1)}\right) \longrightarrow \left(C^\infty(P)[[\lambda]], \{\cdot,\cdot\}^{(2)}\right)$$

is bracket-preserving. This notion of equivalence can be interpreted geometrically as a *gauge action*. Indeed, consider the group

$$\mathscr{G} := \exp\left(\lambda\mathfrak{X}(P)[[\lambda]]\right),$$

whose elements are formal vector fields $X = \sum_{k\geq 1} \lambda^k X_k$ exponentiated via the Lie derivative. Each such element acts on formal multivector fields by

$$\pi \longmapsto \exp(\mathscr{L}_X)\pi,$$

where $\mathscr{L}_X$ denotes the Lie derivative with respect to X. This action preserves the condition $[\pi, \pi] = 0$, and hence restricts to an action on the space of formal Poisson structures. In this sense, two formal Poisson structures are equivalent precisely when they lie in the same orbit of this gauge action.

Moreover, an ordinary Poisson structure π_0 may be deformed to a formal Poisson structure π, in direct analogy with the way in which a star product deforms the product of a commutative algebra: π_0 is identified with the zeroth-order term of the formal power series

$$\sum_{i=0}^{\infty} \lambda^i \pi_i$$

defining π. The set of such formal Poisson structures is denoted by $\mathrm{FPois}_{\pi_0}(P)$ and is defined as the quotient

$$\mathrm{FPois}_{\pi_0}(P) := \left\{\pi \;\middle|\; \pi = \pi_0 + \sum_{i=1}^{\infty} \lambda^i \pi_i\right\} \Big/ \sim,$$

where $\sim$ denotes equivalence under formal diffeomorphisms, as defined above. A formal Poisson structure induces a linear map

$$\pi^\sharp : \Omega^1(P)[[\lambda]] \longrightarrow \mathfrak{X}(P)[[\lambda]],$$

defined by $\alpha \mapsto i_\alpha \pi$, which is used to define the Hamiltonian vector field associated to $f \in C^\infty(P)[[\lambda]]$, namely $\pi^\sharp(df)$.

2.. Formal Symplectic Structures and the Symplectic Case

A distinguished class of formal Poisson structures arises from **formal symplectic forms**. A formal symplectic structure on a manifold P is a formal series of closed 2-forms

$$\omega = \omega_0 + \sum_{k=1}^{\infty} \lambda^k \omega_k \ \in\ \Omega^2(P)[[\lambda]],$$

where ω_0 is an ordinary symplectic form (i.e., non-degenerate and closed). The associated bundle map

$$\omega^\flat : \mathfrak{X}(P)[[\lambda]] \longrightarrow \Omega^1(P)[[\lambda]], \qquad X \longmapsto i_X\omega,$$

is invertible because its zeroth-order term $\omega_0^\flat$ is an isomorphism. Its inverse, denoted by $(\omega^\flat)^{-1}$, defines a formal Poisson bivector

$$\pi_\omega = (\omega^\flat)^{-1} : \Omega^1(P)[[\lambda]] \longrightarrow \mathfrak{X}(P)[[\lambda]],$$

which evidently deforms the Poisson structure π_0 corresponding to ω_0.

Conversely, given a formal Poisson structure $\pi = \pi_0 + \sum_{k\geq 1} \lambda^k \pi_k$ with π_0 symplectic, one can formally invert the map $\pi^\sharp$ to obtain a unique formal symplectic form $\omega = \omega_0 + \sum_{k\geq 1} \lambda^k \omega_k$ such that $\pi = \pi_\omega$. Hence there exists a bijection between formal Poisson deformations of a symplectic structure π_0 and formal symplectic forms deforming its inverse ω_0.

Remark 7. *Under this bijection, two formal Poisson structures are equivalent (via a formal diffeomorphism) if and only if the corresponding formal symplectic forms are cohomologous in the formal de Rham cohomology $H^2_{dR}(P)[[\lambda]]$. Consequently, when π_0 is symplectic, the set of equivalence classes of formal Poisson structures deforming π_0 is isomorphic to the space of formal cohomology classes of degree two:*

$$\mathrm{FPois}_{\pi_0}(P) \;\cong\; \lambda H^2_{dR}(P)[[\lambda]].$$

This result will be used in the following sections.

3.. The Vanishing Zeroth-Order Case

The remainder of this work focuses on the subset $\mathrm{FPois}_0(P)$, where the undeformed Poisson structure is the trivial one, $\pi_0 = 0$. In this situation a formal Poisson structure takes the form

$$\pi = \lambda\pi_1 + \lambda^2\pi_2 + \cdots,$$

so the map $\pi^\sharp$ begins at order λ. As a result, for any closed 2-form $B = \sum_{k\geq 0} \lambda^k B_k \in \Omega^2(P)[[\lambda]]$ the composition $\mathrm{id} + B^\flat\pi^\sharp$ is invertible (its zeroth-order term is the identity), and the gauge transformation $\tau_B(\pi)$ defined in the next section is always well defined. This flexibility makes $\mathrm{FPois}_0(P)$ the natural arena for studying the interplay between Morita equivalence and B-field actions.

II.. Gauge Transformations by Closed 2-Forms (B-Fields)

1.. Definition of gauge transformation.

Consider the following situation. Let $B \in \Omega^2(P)$ be a closed 2-form. In analogy with the bundle map associated with a bivector field, one defines the bundle morphism $B^\flat : TP \longrightarrow T^*P$ by

$$B^\flat(X) := i_X B,$$

for every vector field $X \in \mathfrak{X}(P)$. Equivalently, $B^\flat(X)(Y) = B(X,Y)$, for all vector fields $X, Y \in \mathfrak{X}(P)$. Let now $\pi \in \mathfrak{X}^2(P)$ be a Poisson structure, with associated bundle map

$$\pi^\sharp : T^*P \longrightarrow TP, \qquad \pi^\sharp(\alpha) = i_\alpha \pi.$$

Assume that the bundle endomorphism $\mathrm{id} + B^\flat\pi^\sharp : T^*P \longrightarrow T^*P$ is invertible. One may then define a new bivector field $\tau_B(\pi)$ through the bundle map

$$(\tau_B(\pi))^\sharp := \pi^\sharp\big(\mathrm{id} + B^\flat\pi^\sharp\big)^{-1}.$$

The closedness condition $dB = 0$ ensures that $\tau_B(\pi)$ is again a Poisson structure on P. This construction is called the **gauge transformation** of π by the closed 2-form B, which is usually referred to as a B-**field**. We now illustrate the gauge transformation $\tau_B(\pi)$ defined above with three concrete examples. In each case, $B \in \Omega^2(P)$ is a closed 2-form and π is an ordinary Poisson structure on a manifold P. The map $\mathrm{id} + B^\flat\pi^\sharp$ is assumed to be invertible.

Example 20 (Symplectic manifold: adding a B-field)**.** *Let (P, ω) be a symplectic manifold and let π be the corresponding Poisson bivector (so $\pi^\sharp = \omega^{-1}$). For any closed 2-form B such that $\omega + B$ is non-degenerate, the gauge transformation yields*

$$\tau_B(\pi) \longleftrightarrow \omega + B,$$

i.e., the symplectic form $\omega + B$ corresponds to the Poisson bivector $\tau_B(\pi)$. Indeed, using the defining formula,

$$(\tau_B(\pi))^\sharp = \pi^\sharp(\mathrm{id} + B^\flat\pi^\sharp)^{-1} = \omega^{-1}(\mathrm{id} + B^\flat\omega^{-1})^{-1} = (\omega + B)^{-1},$$

where the last equality follows from the identity $(\omega + B)^{-1} = \omega^{-1}(\mathrm{id} + B^\flat\omega^{-1})^{-1}$. Thus, gauge transformation simply adds the B-field to the symplectic form. This recovers the classical notion of a B-field shift in symplectic geometry.

Example 21 (Constant B-field on a two-dimensional Poisson manifold)**.** *Let $P = \mathbb{R}^2$ with coordinates (x, y). Consider the Poisson bivector*

$$\pi = f(x,y)\,\frac{\partial}{\partial x} \wedge \frac{\partial}{\partial y},$$

Swhere f is a smooth function. The corresponding bundle map is

$$\pi^\sharp(dx) = f\,\frac{\partial}{\partial y}, \qquad \pi^\sharp(dy) = -f\,\frac{\partial}{\partial x}.$$

Take a constant closed 2-form $B = c\,dx \wedge dy$ with $c \in \mathbb{R}$. Then $B^\flat\pi^\sharp : T^*P \to T^*P$ acts by

$$B^\flat\pi^\sharp(dx) = B^\flat(f\partial_y) = f\,B(\partial_y, \cdot) = -cf\,dx,$$

$$B^\flat\pi^\sharp(dy) = B^\flat(-f\partial_x) = -f\,B(\partial_x, \cdot) = -cf\,dy.$$

Hence $\mathrm{id} + B^\flat\pi^\sharp$ is diagonal in the basis $\{dx, dy\}$ with eigenvalue $1 - cf$ (taking into account the sign; a careful calculation gives $(\mathrm{id} + B^\flat\pi^\sharp)(dx) = (1 - cf)dx$ and similarly for dy). Provided $1 - cf \neq 0$ everywhere, the inverse exists and

$$(\tau_B(\pi))^\sharp(dx) = \pi^\sharp\big((\mathrm{id} + B^\flat\pi^\sharp)^{-1}dx\big) = \pi^\sharp\big((1 - cf)^{-1}dx\big) = \frac{f}{1 - cf}\,\frac{\partial}{\partial y},$$

and analogously for dy. Consequently,

$$\tau_B(\pi) = \frac{f}{1 - cf}\,\frac{\partial}{\partial x} \wedge \frac{\partial}{\partial y}.$$

Thus the gauge transformation rescales the Poisson structure pointwise by the factor $\frac{f}{1-cf}$. This example shows that a constant B-field can modify the "strength" of the Poisson bracket in a position-dependent way, potentially creating singularities when $1 - cf$ vanishes.

Example 22 (Trivial Poisson structure)**.** *Let $\pi = 0$ be the zero Poisson bivector on any manifold P. Then $\pi^\sharp = 0$, so $\mathrm{id} + B^\flat\pi^\sharp = \mathrm{id}$ is trivially invertible. The formula gives*

$$(\tau_B(0))^\sharp = 0 \circ (\mathrm{id})^{-1} = 0,$$

hence $\tau_B(0) = 0$ for every closed B-field. This example demonstrates that gauge transformations have no effect on a vanishing Poisson structure, which is consistent with the fact that there are no symplectic leaves to be modified.

The lesson is that, in Example 1, a B-field is added to the symplectic form on the leaves; in Example 2, the Poisson bivector undergoes a rescaling; and, in Example 3, the action on the zero Poisson structure is trivial. For these reasons, the examples are of particular significance, as they illustrate several noteworthy features of gauge transformations.

2.. Effect on symplectic leaves and Poisson cohomology

In this subsection, I expect to show that gauge transformations by closed 2-forms preserve the underlying characteristic distribution of the Poisson structure, whilst modifying the symplectic forms on its leaves. With that being said, since

$$\eta := (\mathrm{id} + B^\flat \pi^\sharp)^{-1}$$

is an invertible bundle endomorphism of T^*P, one has

$$\mathrm{Im}\big((\tau_B(\pi))^\sharp\big) = \mathrm{Im}\big(\pi^\sharp \circ \eta\big) = \mathrm{Im}(\pi^\sharp).$$

Consequently, the characteristic distributions of π and $\tau_B(\pi)$ coincide. Since the symplectic foliation of a Poisson manifold is determined by the image of the anchor map $\pi^\sharp$, it follows that π and $\tau_B(\pi)$ possess the same symplectic leaves. Although the symplectic foliation remains unchanged, the symplectic structures carried by the leaves are modified by the B-field. The distribution induced by the Poisson structure is integrable in the Stefan–Sussmann sense, so it partitions P into symplectic leaves, which we'll simply call leaves of π. Let

$$(S, \omega_S)$$

be a symplectic leaf of π. The symplectic form induced on S by the gauge-transformed Poisson structure $\tau_B(\pi)$ is given by

$$\omega_S^B = \omega_S + \iota_S^* B,$$

where

$$\iota_S : S \hookrightarrow P$$

denotes the canonical inclusion. Thus, gauge transformations preserve the underlying foliation whilst altering the leafwise symplectic geometry. In particular, the effect of the B-field is entirely encoded by its restriction to the individual symplectic leaves. From this perspective, gauge transformations may be viewed as a mechanism for modifying Poisson structures without changing their characteristic distribution. Gauge-equivalent Poisson structures also possess closely related Poisson cohomologies. Indeed, the cotangent Lie algebroids associated with π and $\tau_B(\pi)$ are naturally isomorphic, and consequently many cohomological invariants are preserved under gauge transformation. This observation is particularly important in the study of Morita equivalence, where gauge-equivalent Poisson structures frequently determine equivalent geometric data.

3.. Gauge equivalence of formal Poisson structures

Our next step is to extend the notion of gauge transformation to formal Poisson structures and to study the corresponding notion of equivalence. In the formal setting, the B-fields are likewise taken to be formal power series

$$B = \sum_{i=0}^{\infty} \lambda^i B_i \in \Omega^2(P)[[\lambda]],$$

where each B_i is a closed 2-form on P. In complete analogy with the ordinary case, one considers the $\mathbb{K}[[\lambda]]$-linear map

$$\mathrm{id} + B^\flat \pi^\sharp \; : \; \Omega^1(P)[[\lambda]] \longrightarrow \Omega^1(P)[[\lambda]].$$

As we did for the non-formal scenenario, we now illustrate the notion of gauge transformation $\tau_B(\pi)$ with several concrete examples. In each case, $B = \sum_{k\geq 0} \lambda^k B_k$ is a closed formal 2-form, and $\pi = \sum_{k\geq 0} \lambda^k \pi_k$ is a formal Poisson structure. Recall that when $\pi_0 = 0$ (the case of primary interest in this work) the map $\mathrm{id} + B^\flat \pi^\sharp$ is automatically invertible, so $\tau_B(\pi)$ is defined for every closed B.

Example 23 (Constant B-field on $\mathbb{R}^2$)**.** *Let $P = \mathbb{R}^2$ with coordinates (x, y). Consider the formal Poisson structure*

$$\pi = \lambda \, \frac{\partial}{\partial x} \wedge \frac{\partial}{\partial y},$$

so that $\pi_0 = 0$ and $\pi_1 = \partial_x \wedge \partial_y$. Let $B = b\, dx \wedge dy$ be a constant (closed) 2-form, where $b = b_0 + \lambda b_1 + \cdots$ is a formal constant (i.e., a series in λ with real coefficients). A direct computation using the formula $(\tau_B(\pi))^\sharp = \pi^\sharp (\mathrm{id} + B^\flat \pi^\sharp)^{-1}$ yields

$$\tau_B(\pi) = \frac{\lambda}{1 + \lambda b} \, \frac{\partial}{\partial x} \wedge \frac{\partial}{\partial y},$$

where the fraction is understood as a formal power series: $\frac{\lambda}{1+\lambda b} = \lambda - \lambda^2 b + \lambda^3 b^2 - \cdots$. Thus the gauge transformation simply rescales the constant Poisson bivector by a formal factor. In particular, if b is chosen so that $1 + \lambda b$ is a unit in $\mathbb{R}[[\lambda]]$ (which is always the case), we obtain a family of inequivalent formal Poisson structures parametrised by the cohomology class $[B] = b\,[dx \wedge dy]$.

Example 24 (Symplectic case: gauge transformation as addition of B)**.** *Suppose π_0 is a genuine symplectic structure on P with symplectic form ω_0 (i.e., $\pi_0 = \omega_0^{-1}$). Let $\pi = \pi_0 + \lambda\pi_1 + \cdots$ be any formal Poisson structure deforming π_0, and let $\omega = \omega_0 + \lambda\omega_1 + \cdots$ be the corresponding formal symplectic form (obtained by formal inversion). For a closed 2-form $B = B_0 + \lambda B_1 + \cdots$ with B_0 sufficiently small (so that $\omega_0 + B_0$ remains non-degenerate), the gauge transformation $\tau_B(\pi)$ corresponds, under the symplectic-Poisson correspondence, to the formal symplectic form $\omega + B$. In other words,*

$$\tau_B(\pi) \longleftrightarrow \omega + B.$$

This is a direct consequence of Lemma 2.8 in [1]. For the undeformed Poisson structure $\pi = \pi_0$ (i.e., $\pi_k = 0$ for $k \geq 1$), we obtain

$$\tau_B(\pi_0) \longleftrightarrow \omega_0 + B,$$

so the gauge transformation adds the B-field to the symplectic form. This recovers the classical gauge transformation of a symplectic structure.

Example 25 (Morita equivalence on the torus T^2)**.** *Let $P = T^2 = \mathbb{R}^2/\mathbb{Z}^2$ with angular coordinates (θ, ϕ). Consider the formal Poisson structure*

$$\pi = \lambda \frac{\partial}{\partial\theta} \wedge \frac{\partial}{\partial\phi},$$

which is the translation-invariant Poisson structure of rank two (the formal analogue of the standard symplectic structure on the torus). Take a B-field of the form

$$B = m\, d\theta \wedge d\phi,$$

where $m = m_0 + \lambda m_1 + \cdots$ is a formal constant. The gauge transformation gives

$$\tau_B(\pi) = \frac{\lambda}{1 + \lambda m} \frac{\partial}{\partial\theta} \wedge \frac{\partial}{\partial\phi}.$$

Now suppose that m_0 is an integer (e.g., $m_0 = 1$). Then B corresponds to an integral cohomology class in $H^2_{dR}(T^2)[[\lambda]]$ (with the understanding that the constant term lies in $2\pi i H^2(T^2, \mathbb{Z})$ if we work over $\mathbb{C}$). According to the main theorem (Theorem 1.1), two formal Poisson structures of this form are Morita equivalent if and only if their classes in $\mathrm{FPois}_0(T^2)$ differ by the action of a diffeomorphism and a gauge transformation. Since the diffeomorphism group of the torus acts by linear transformations on the bivector, we see that, for instance, π and $\tau_{d\theta\wedge d\phi}(\pi)$ are Morita equivalent. In fact, the integer m_0 parametrises Morita equivalence classes of such formal Poisson structures, in direct analogy with the classification of Morita equivalence classes of non-commutative tori.

In summary, these examples, much like those presented in Section 6.1, are of interest both from a pedagogical perspective and because they further emphasise several important features of gauge transformations. More precisely, in Example 4, the Poisson structure undergoes a rescaling, thereby altering its effective "strength"; Example 5 shows that, in the non-degenerate setting, such transformations amount to the addition of a closed 2-form to the symplectic structure; finally, Example 6 illustrates how gauge transformations may produce Morita-equivalent Poisson structures, especially when the B-field possesses integral periods.

4.. Relation to symplectic groupoids

Gauge transformations also admit a natural interpretation at the level of symplectic groupoids. Recall that an integrable Poisson manifold (P, π) may be integrated by a symplectic groupoid $(\mathcal{G} \rightrightarrows P, \omega)$, whose symplectic structure encodes the Poisson geometry of P in the sense that the Poisson bivector $\pi \in \mathfrak{X}^2(P)$ coincides with the Poisson structure induced on the base manifold P by the multiplicative symplectic form $\omega \in \Omega^2(\mathcal{G})$. Under suitable assumptions (e.g., when $\mathcal{G}$ is source-simply-connected and B is a closed 2-form on P), gauge transformations of π correspond to canonical modifications of the symplectic form on $\mathcal{G}$.

In particular, the relation extends naturally to the gauge-transformed bivector $\tau_B(\pi) \in \mathfrak{X}^2(P)$ so that $(\mathcal{G} \rightrightarrows P, \omega_B)$ integrates $(P, \tau_B(\pi))$, where

$$\omega_B := \omega + t^*B - s^*B.$$

Thus, gauge transformations on the base Poisson manifold induce corresponding modifications of the symplectic structure on the integrating groupoid, whilst preserving the underlying Lie groupoid structure.

III.. Morita Equivalence of Poisson Manifolds

1.. Symplectic bimodules (double symplectic realisations)

A fundamental notion in Poisson geometry is that of a *symplectic realisation*, i.e., a Poisson map from a symplectic manifold onto a Poisson manifold. Symplectic bimodules, also called *double symplectic realisations* or *dual pairs*, generalise this idea by providing a symplectic manifold that simultaneously realises two Poisson manifolds.

Definition 25 (Symplectic bimodule)**.** *Let (P_1, π_1) and (P_2, π_2) be Poisson manifolds. A* symplectic bimodule *from P_1 to P_2 is a diagram*

$$(P_1, \pi_1) \xleftarrow{J_1} (S, \omega) \xrightarrow{J_2} (P_2, \pi_2),$$

where (S, ω) is a symplectic manifold, $J_1 : S \to P_1$ is a Poisson map, $J_2 : S \to P_2$ is an anti-Poisson map, and the subalgebras $J_1^* C^\infty(P_1)$ and $J_2^* C^\infty(P_2)$ Poisson commute in $C^\infty(S)$.

The condition that the images of the two pullbacks Poisson commute is equivalent to requiring that

$$\{J_1^* f, J_2^* g\}_\omega = 0 \quad \text{for all } f \in C^\infty(P_1),\ g \in C^\infty(P_2).$$

This ensures that the two Poisson structures are "opposite" in a suitable sense. The term *dual pair* is also commonly used in the literature; a symplectic bimodule is a dual pair (S, ω, J_1, J_2) with a choice of which map is Poisson and which is anti-Poisson.

For a symplectic bimodule to be useful for Morita theory, one imposes additional regularity conditions.

Definition 26 (Equivalence bimodule)**.** *A symplectic bimodule*

$$(P_1, \pi_1) \xleftarrow{J_1} (S, \omega) \xrightarrow{J_2} (P_2, \pi_2)$$

is called an equivalence bimodule *if the maps J_1 and J_2 are surjective submersions, complete (in the sense of Definition 2.10), have connected and simply connected fibres, and the subbundles tangent to their fibres are symplectic orthogonal complements of each other.*

The condition that the fibre tangent distributions are symplectic orthogonal complements means that

$$\ker T J_1 = (\ker T J_2)^\omega.$$

In particular, this implies that the Poisson subalgebras $J_1^* C^\infty(P_1)$ and $J_2^* C^\infty(P_2)$ are commutants of one another in $C^\infty(S)$. Equivalence bimodules are precisely the geometric analogues of the bimodules that establish Morita equivalence of algebras.

2.. Morita equivalence as an equivalence relation

With the notion of equivalence bimodules in place, one can define an equivalence relation on the class of Poisson manifolds.

Definition 27 (Morita equivalence)**.** *Two Poisson manifolds (P_1, π_1) and (P_2, π_2) are* Morita equivalent *if there exists an equivalence bimodule*

$$(P_1, \pi_1) \xleftarrow{J_1} (S, \omega) \xrightarrow{J_2} (P_2, \pi_2).$$

It is natural to ask whether this definition indeed yields an equivalence relation. This is not automatic for arbitrary Poisson manifolds; the existence of equivalence bimodules imposes strong integrability conditions.

Proposition 7. *Morita equivalence defines an equivalence relation on the class of* integrable *Poisson manifolds.*

The key point is that for a Poisson manifold to participate in an equivalence bimodule, its underlying Poisson structure must be integrable, i.e., it must admit a symplectic groupoid integration. Within the class of integrable Poisson manifolds, one can prove:

- *Reflexivity:* Every integrable Poisson manifold is Morita equivalent to itself via the identity bimodule, where $S = P$ and ω is a symplectic realisation of π (the existence of such a realisation is equivalent to integrability).
- *Symmetry:* If (P_1, π_1) and (P_2, π_2) are Morita equivalent via a bimodule S, then the same symplectic manifold S with the roles of the two maps interchanged gives a Morita equivalence in the opposite direction.
- *Transitivity:* Given equivalence bimodules between P_1 and P_2, and between P_2 and P_3, one can form a composition of the two bimodules, known as the *Morita product* or *fibre product*, yielding an equivalence bimodule between P_1 and P_3.

The transitivity property is the most subtle and relies heavily on the integrability assumption.

3.. Relation to symplectic groupoid equivalence

There is a profound connection between Morita equivalence of Poisson manifolds and equivalence of their associated symplectic groupoids. Recall that to any integrable Poisson manifold (P, π), one can associate a source-simply-connected symplectic groupoid $(\Sigma(P), \omega)$ integrating it.

Theorem 5. *Let (P_1, π_1) and (P_2, π_2) be integrable Poisson manifolds with source-simply-connected symplectic groupoids $\Sigma(P_1)$ and $\Sigma(P_2)$. Then (P_1, π_1) and (P_2, π_2) are Morita equivalent if and only if their symplectic groupoids are equivalent in the sense of Lie groupoids.*

The notion of Lie groupoid equivalence is the natural categorical notion of Morita equivalence for Lie groupoids, defined by the existence of a principal bibundle. The theorem establishes a precise correspondence:

$$\boxed{\text{Poisson manifolds up to Morita equivalence}} \longleftrightarrow \boxed{\text{Symplectic groupoids up to groupoid equivalence}}.$$

This correspondence is extremely powerful because groupoid equivalence is a well-understood concept in Lie theory and can often be studied via cohomological and homotopical methods.

4.. Morita invariance of key objects

A fundamental principle in Morita theory is that certain objects and structures are invariant under the equivalence relation. For Poisson manifolds, many important geometric and algebraic invariants are Morita invariant.

Proposition 8. *The following objects are invariant under Morita equivalence of Poisson manifolds:*

1. *The leaf space $P/\sim$ of the symplectic foliation, up to homeomorphism.*
2. *The isomorphism class of the isotropy Lie algebras at corresponding points.*
3. *The Poisson cohomology ring $H^\bullet_\pi(P)$.*
4. *The modular class, an invariant of the Poisson structure up to a certain cohomology class.*
5. *The monodromy groups of the symplectic foliation.*

These invariance results will be explored in more detail in the next section. The underlying reason for this invariance is that Morita equivalent Poisson manifolds have equivalent symplectic groupoids, and all the structures listed above can be reconstructed from the symplectic groupoid in a Morita-invariant manner. For instance, the Poisson cohomology of P is isomorphic to a certain groupoid cohomology of $\Sigma(P)$, which is invariant under groupoid equivalence.

IV.. Morita Invariants

1.. Leaf space homeomorphism

The symplectic foliation of a Poisson manifold is its fundamental geometric structure. Morita equivalence preserves the leaf space as a topological space.

Proposition 9. *Let (P_1, π_1) and (P_2, π_2) be Morita equivalent Poisson manifolds. Then there is a homeomorphism*

$$P_1/\sim_{\pi_1} \cong P_2/\sim_{\pi_2}$$

between their leaf spaces.

The proof relies on the fact that an equivalence bimodule (S, ω, J_1, J_2) provides a correspondence between leaves: the fibres of J_1 and J_2 are symplectic orthogonal complements, which implies that the leaves of P_1 and P_2 are in bijection via the induced maps on the leaf spaces. Moreover, this bijection is a homeomorphism due to the surjective submersion property.

2.. Poisson cohomology rings

The Poisson cohomology $H^\bullet_\pi(P)$ is a graded ring associated to a Poisson manifold. It is a fundamental invariant in deformation theory and Poisson geometry.

Theorem 6 (Morita invariance of Poisson cohomology)**.** *If (P_1, π_1) and (P_2, π_2) are Morita equivalent Poisson manifolds, then there is an isomorphism of graded rings*

$$H^\bullet_{\pi_1}(P_1) \cong H^\bullet_{\pi_2}(P_2).$$

The proof of this result is subtle. It uses the fact that Morita equivalence induces an equivalence of the associated Poisson algebras $C^\infty(P_1)$ and $C^\infty(P_2)$ in a suitable bicategory. From this algebraic equivalence, one obtains an isomorphism of the corresponding Poisson cohomology complexes. Alternatively, in the integrable case, one can show that the Poisson cohomology of P is isomorphic to the differentiable cohomology of its source-simply-connected symplectic groupoid, and this groupoid cohomology is Morita invariant.

3.. Modular class

The modular class of a Poisson manifold is a first Poisson cohomology class that measures the failure of the Poisson structure to preserve a volume form. It is an obstruction to the existence of a unimodular Poisson structure.

Definition 28. *Let (P, π) be a Poisson manifold with a chosen volume form μ. The modular vector field is defined by*

$$X_\mu := \operatorname{div}_\mu(\pi^\sharp).$$

The modular class is the cohomology class $[X_\mu] \in H^1_\pi(P)$, which is independent of the choice of μ.

The modular class is a Morita invariant.

Proposition 10. *If (P_1, π_1) and (P_2, π_2) are Morita equivalent Poisson manifolds, then their modular classes correspond under the isomorphism of Poisson cohomology:*

$$\operatorname{mod}(P_1) \longleftrightarrow \operatorname{mod}(P_2).$$

This is a consequence of the Morita invariance of Poisson cohomology and the fact that the modular class can be defined in a purely algebraic way that is invariant under the appropriate equivalence of Poisson algebras.

4.. Monodromy groups

The monodromy of a Poisson manifold is a group associated to its symplectic foliation. It measures the obstruction to the existence of a global symplectic realisation.

Definition 29. *For a Poisson manifold (P, π), the monodromy group at a leaf L is the image of the homomorphism*

$$\pi_1(L) \to \operatorname{Aut}(\mathfrak{g}_L)$$

given by the action of the fundamental group of the leaf on the isotropy Lie algebra $\mathfrak{g}_L$ via the linearised Poisson structure.

The monodromy groups are Morita invariant.

Proposition 11. *Under Morita equivalence, the isomorphism class of the monodromy groups at corresponding leaves is preserved. In particular, if (P_1, π_1) and (P_2, π_2) are Morita equivalent, there is a bijection between their leaf spaces such that the monodromy groups at corresponding leaves are isomorphic.*

This invariance follows from the fact that the symplectic groupoid of an integrable Poisson manifold contains all the monodromy information, and groupoid equivalence preserves the isotropy groups at corresponding points.

5.. Isotropy Lie algebras

The isotropy Lie algebra $\mathfrak{g}_x$ at a point $x \in P$ is the Lie algebra of the isotropy group of the symplectic groupoid, or equivalently, the Lie algebra of the symplectic leaf at x.

Proposition 12. *If (P_1, π_1) and (P_2, π_2) are Morita equivalent, then for every $x \in P_1$ there exists $y \in P_2$ such that the isotropy Lie algebras are isomorphic:*

$$\mathfrak{g}_x \cong \mathfrak{g}_y.$$

Moreover, the bijection between leaves induced by the Morita equivalence respects the isotropy Lie algebras in the sense that there is an isomorphism of Lie algebra bundles over the leaf space. This is a manifestation of the fact that Morita equivalence of Poisson manifolds corresponds precisely to an equivalence of their associated Lie algebroids, where the isotropy Lie algebras are the fibres of the algebroid.

Remark 8. *The Morita invariants discussed above are not exhaustive. Other invariants include the fundamental group of the leaves, the leaf-wise symplectic cohomology, and the Picard group of the Poisson manifold. The study of Morita invariants is an active area of research in Poisson geometry.*

Chapter 3

Deformation of Poisson Morphisms

I.. Cohomology and Existence of Deformations

The study of deformations of Poisson morphisms is a central component in understanding how Morita equivalence behaves under formal deformations. Given a fixed Poisson morphism between two Poisson algebras, and given formal deformations of the Poisson structures on the domain and codomain, we seek to deform the morphism itself into a Poisson morphism with respect to the deformed structures. This section presents a systematic cohomological approach to this deformation problem, with the statement and proof of three propositions that justify the existence and uniqueness of such deformations of the Morita equivalence.

Let $\mathscr{A}$ and $\mathscr{B}$ be commutative algebras over a field $\mathbb{K}$ of characteristic zero, equipped with Poisson brackets $\pi_0 = \{\cdot,\cdot\}_{\mathscr{A}}$ and $\sigma_0 = \{\cdot,\cdot\}_{\mathscr{B}}$, respectively. Fix a Poisson morphism

$$\phi_0 : (\mathscr{A}, \pi_0) \to (\mathscr{B}, \sigma_0).$$

At its core, this is just an ordinary formal deformation problem and the cohomology that controls this problem is a variant of Chevalley–Eilenberg cohomology adapted to the Leibniz rule of Poisson brackets. Consider the space of k-cochains

$$C^k_{\mathrm{CE}}(\mathscr{A}, \mathscr{B}) := \left\{ D : \mathscr{A}^{\times k} \to \mathscr{B} \mid D \text{ is multilinear and antisymmetric} \right\},$$

with differential δ defined by

$$\begin{aligned}(\delta D)(a_0, \dots, a_k) = &\sum_{j=0}^{k} (-1)^j \{\phi_0(a_j), D(a_0, \dots, \widehat{a_j}, \dots, a_k)\}_{\mathscr{B}} \\ &+ \sum_{i<j} (-1)^{i+j} D(\{a_i, a_j\}_{\mathscr{A}}, a_0, \dots, \widehat{a_i}, \dots, \widehat{a_j}, \dots, a_k).\end{aligned}$$

This is the standard Chevalley–Eilenberg differential for the Lie algebra $(\mathscr{A}, \pi_0)$ acting on $\mathscr{B}$ via $a \mapsto \{\phi_0(a), \cdot\}_{\mathscr{B}}$. However, since Poisson brackets satisfy the Leibniz rule with respect to the associative product, we must restrict to a subcomplex of *multiderivations along* ϕ_0. We define

$$C^k_{\mathrm{CE,der}}(\mathscr{A}, \mathscr{B}) \subset C^k_{\mathrm{CE}}(\mathscr{A}, \mathscr{B})$$

as the subspace of cochains D satisfying, for each $l = 1, \dots, k$,

$$D(a_1, \dots, a_l a_l', \dots, a_k) = \phi_0(a_l) D(a_1, \dots, a_l', \dots, a_k) + D(a_1, \dots, a_l, \dots, a_k)\phi_0(a_l').$$

The differential δ preserves this subspace, giving a subcomplex $(C^\bullet_{\mathrm{CE,der}}(\mathscr{A}, \mathscr{B}), \delta)$. Its cohomology is denoted by $H^\bullet_{\mathrm{CE,der}}(\mathscr{A}, \mathscr{B})$. The key insight is that obstructions to deforming Poisson morphisms live in the second cohomology group of this complex.

The existence of deformations is governed by the vanishing of the second cohomology group $H^2_{\mathrm{CE,der}}(\mathscr{A}, \mathscr{B})$. The construction proceeds inductively, with the cohomology class of the obstruction at each order determining whether the deformation can be extended to the next order.

Proposition 13 (Existence of deformations)**.** *Assume there exists a horizontal lift along ϕ_0, i.e., an $\mathscr{A}$-linear map*

$$h:\ C^1_{\mathrm{CE,der}}(\mathscr{A},\mathscr{B}) \to \mathrm{Der}(\mathscr{B})$$

such that $\phi_0^* h(D) = D$ for all D. If $H^2_{\mathrm{CE,der}}(\mathscr{A},\mathscr{B}) = \{0\}$, then there exists a derivation $X \in \lambda\,\mathrm{Der}(\mathscr{B})[[\lambda]]$ such that $\Phi = \exp(X)\phi_0$ solves the deformation problem.

Remark 9. *We denote by $X_{(k)}$ the k-th order partial sum of X.*

Proof. The proof is constructed inductively. For the base case, we define $X_{(0)} = 0$, so that $\exp(X_{(0)})\phi_0 = \phi_0$ that, by hypothesis, is a Poisson morphism. For the induction hypothesis, we suppose we have found a formal derivation X, such that $\exp(X_{(k)})\phi_0$ is a Poisson morphism up to order k. By definition, a poisson morphism $\Phi : \mathscr{A} \to \mathscr{B}$ preserves the poisson structure σ on $\mathscr{B}$. By the induction hypothesis, this implies that

$$\sigma(\Phi_{(k)}\phi_0(a_1), \Phi_{(k)}\phi_0(a_2)) - \Phi_{(k)}\phi_0(\pi(a_1,a_2)) = \text{higher order residue} \tag{3.1}$$

Which means:

$$\Phi^{-1}_{(k)}\sigma(\Phi_{(k)}\phi_0(a_1), \Phi_{(k)}\phi_0(a_2)) = \phi_0(\pi(a_1,a_2)) + \lambda^{k+1}R_{k+1}(a_1,a_2) + \dots \tag{3.2}$$

where $\Phi_{(k)} = \exp(X_{(k)})$ and $R_{k+1} : \mathscr{A}\times\mathscr{A} \to \mathscr{B}$ is a 2-cocyle in $C^\bullet_{CE,der}(\mathscr{A},\mathscr{B})$ and, equally important, the error term of order $k+1$.

Consider $X_{k+1} \in \mathrm{Der}(B)$ and define $\Phi_{(k+1)} := \Phi_{(k)}\exp(\lambda^{k+1}X_{k+1}) = \Phi_{(k)}\Phi_{k+1}$. We then compute the equivalent of equation (3.2), where we defined the error term, for order $k+1$:

$$\begin{aligned}
\Phi^{-1}_{(k+1)}\sigma(\Phi_{k+1}\phi_0(a_1), \Phi_{(k+1)}\phi_0(a_2)) &= \Phi^{-1}_{k+1}\Phi^{-1}_{(k)}\sigma(\Phi_{(k)}\Phi_{k+1}\phi_0(a_1), \Phi_{(k)}\Phi_{k+1}\phi_0(a_2)) \\
&= \phi_0(\pi(a_1,a_2)) + \lambda^{k+1}R_{k+1}(a_1,a_2) - \lambda^{k+1}X_{k+1}\phi_0\{a_1,a_2\}_{\mathscr{A}} + \dots \\
&\quad + \lambda^{k+1}\{\phi_0(a_1), X_{k+1}\phi_0(a_2)\}_{\mathscr{B}} + \lambda^{k+1}\{X_{k+1}\phi_0(a_1), \phi_0(a_2)\}_{\mathscr{B}} + \dots
\end{aligned} \tag{3.3}$$

Since the error must be zero at $k+1$ order for $\Phi_{(k+1)}$ to an Poisson morphism (up to this order), from (3.3) we get:

$$\begin{aligned}
0 = \lambda^{k+1}\big(&R_{k+1}(a_1,a_2) - \phi_0\{a_1,a_2\}_{\mathscr{A}} \\
&+ \{\phi_0(a_1), X_{k+1}\phi_0(a_2)\}_{\mathscr{B}} + \{X_{k+1}\phi_0(a_1), \phi_0(a_2)\}_{\mathscr{B}}\big)
\end{aligned} \tag{3.4}$$

Which implies

$$-R_{k+1} = \delta\phi_0^* X_{k+1}. \tag{3.5}$$

Since, by hypothesis, $H^2_{CE,der}(\mathscr{A},\mathscr{B}) = \{0\}$, every 2-cocycle is also a 2-cochain. As R_{k+1} is a 2-cochain, then $\exists\xi_{k+1} \in C^1_{CE,der}(\mathscr{A},\mathscr{B})$ 1-cocycle such that $-R_{k+1} = \delta\xi_{k+1}$. Hence, any horizontal lift $X_{k+1} := \xi^h_{k+1}$ of ξ_{k+1} satisfies (3.5) and, therefore, is a Poisson morphism in the order $k+1$. □

In this proof, we used the following lemma:

Lemma I..1. *The map $R_{k+1} : \mathscr{A}\times\mathscr{A} \to \mathscr{B}$ defined on 3.2 is a 2-cocyle in $C^\bullet_{CE,der}(\mathscr{A},\mathscr{B})$*

Proof. We show that the obstruction R_{k+1} to extending ϕ_0 to order $k+1$ is a Chevalley–Eilenberg 2-cocycle, i.e. that $\delta R_{k+1} = 0$. Throughout,

$$\sum_{\circlearrowleft}$$

denotes summation over the three cyclic permutations (a_i,a_j,a_k) of a fixed triple $(a_1,a_2,a_3) \in A$. Fixing an ordering of a_1,a_2,a_3 and expanding $\Phi_{(k)}$, $\Phi^{-1}_{(k)}$ in powers of λ up to order $k+1$ gives

$$\begin{aligned}
\Phi^{-1}_{(k)}\sigma\Big(\Phi_{(k)}\phi_0(a_1),\ \sigma\big(\Phi_{(k)}\phi_0(a_2), \Phi_{(k)}\phi_0(a_3)\big)\Big) &= \phi_0\big(\pi(a_1,\pi(a_2,a_3))\big) + \lambda^{k+1}R_{k+1}\big(a_1,\pi(a_2,a_3)\big) + \cdots \\
&\quad + \Phi^{-1}_{(k)}\sigma\Big(\Phi_{(k)}\phi_0(a_1),\ \Phi_{(k)}\big(\lambda^{k+1}R_{k+1}(a_2,a_3) + \cdots\big)\Big).
\end{aligned}$$

Performing the same expansion for the two remaining cyclic orderings and summing all three identities, the left-hand side becomes

$$\begin{aligned}
\mathrm{LHS} &= \Phi^{-1}_{(k)}\sigma\Big(\Phi_{(k)}\phi_0(a_1),\ \sigma\big(\Phi_{(k)}\phi_0(a_2), \Phi_{(k)}\phi_0(a_3)\big)\Big) \\
&\quad + \Phi^{-1}_{(k)}\sigma\Big(\Phi_{(k)}\phi_0(a_2),\ \sigma\big(\Phi_{(k)}\phi_0(a_3), \Phi_{(k)}\phi_0(a_1)\big)\Big) \\
&\quad + \Phi^{-1}_{(k)}\sigma\Big(\Phi_{(k)}\phi_0(a_3),\ \sigma\big(\Phi_{(k)}\phi_0(a_1), \Phi_{(k)}\phi_0(a_2)\big)\Big) \\
&= \Phi^{-1}_{(k)}\left[\sum_{\circlearrowleft}\sigma\Big(\Phi_{(k)}\phi_0(a_i),\ \sigma\big(\Phi_{(k)}\phi_0(a_j), \Phi_{(k)}\phi_0(a_k)\big)\Big)\right],
\end{aligned}$$

using linearity of $\Phi_{(k)}^{-1}$ in the last step.

Since σ satisfies the Leibniz identity (it is a Poisson structure) and $\Phi_{(k)}$ is an automorphism (hence preserves this identity), the bracketed cyclic sum vanishes identically:

$$\sum_{\circlearrowleft} \sigma\Big(\Phi_{(k)}\phi_0(a_i),\, \sigma\big(\Phi_{(k)}\phi_0(a_j), \Phi_{(k)}\phi_0(a_k)\big)\Big) = 0.$$

As $\Phi_{(k)}^{-1}$ is linear, $\Phi_{(k)}^{-1}(0) = 0$, and therefore

$$\text{LHS} = 0.$$

By the same reasoning applied to π — using that π satisfies the Leibniz identity and that ϕ_0, being a Poisson morphism, is linear — the leading term of the expansion satisfies

$$\phi_0\left(\sum_{\circlearrowleft} \pi\big(a_i, \pi(a_j, a_k)\big)\right) = \phi_0(0) = 0.$$

By Step 2 the full cyclic sum vanishes, and by Step 3 its leading (λ^0) term vanishes as well; hence the remaining terms of the expansion must cancel cyclically order by order in λ. Moreover, since $\Phi_{(k)}$ and $\Phi_{(k)}^{-1}$ act as the identity at order 0 in λ, any contribution from their higher-order parts is pushed to order λ^{k+2} or beyond and may be discarded. Extracting the coefficient of λ^{k+1} therefore yields

$$\sum_{\circlearrowleft} R_{k+1}\big(a_i, \pi(a_j, a_k)\big) \;+\; \sum_{\circlearrowleft} \sigma\big(\phi_0(a_i),\, R_{k+1}(a_j, a_k)\big) \;=\; 0.$$

Writing $\sigma(\cdot,\cdot) = \{\cdot,\cdot\}_B$ and $\pi(\cdot,\cdot) = \{\cdot,\cdot\}_A$, the equation above reads

$$\sum_{\circlearrowleft} R_{k+1}\big(a_i, \{a_j, a_k\}_A\big) \;+\; \sum_{\circlearrowleft} \big\{\phi_0(a_i),\, R_{k+1}(a_j, a_k)\big\}_B \;=\; 0,$$

that is, $\delta R_{k+1} = 0$. Hence R_{k+1} is a 2-cocycle in $C^{\bullet}_{\mathrm{CE}}(A, B)$. □

Proposition 14 (Uniqueness of deformations)**.** *Assume that $H^1_{\mathrm{CE,der}}(\mathscr{A}, \mathscr{B}) = \{0\}$. If $X, \overline{X} \in \lambda\,\mathrm{Der}(\mathscr{B})[[\lambda]]$ are two solutions to the deformation problem, then there exist a vertical derivation $V \in \lambda\,\mathrm{Der}(\mathscr{B})[[\lambda]]$ (i.e., $V \circ \phi_0 = 0$) and a Hamiltonian derivation $X_H = \sigma(H, \cdot)$ with $H \in \lambda\mathscr{B}[[\lambda]]$ such that*

$$\exp(\overline{X}) = \exp(X_H)\exp(X)\exp(V).$$

In particular, any two solutions are equivalent in the natural sense.

Proof. The proof is, again, by induction. For $k = 0$, we define $Y_{(0)} = V_{(0)} = 0$, so that $\exp(Y_{(0)})\exp(X)\exp(V_{(0)}) = \exp(X) = \exp(\overline{X})$, which triavilly implies that they agree. For the induction hypothesis, we suppose that there exists some $Y_{(k)}, V_{(k)} \in \lambda\,\mathrm{D}(\mathscr{B})[[\lambda]]$, with $V_{(k)}$ vertical and $Y_{(k)}$ a derivation of σ, such that

$$\exp(Y_{(k)})\exp(X)\exp(V_{(k)}) = \exp(\overline{X}) + \textbf{higher order residue} \tag{3.6}$$

From (3.6) we get

$$\exp(Y_{(k)})\exp(X)\exp(V_{(k)})\exp(-\overline{X}) = \mathrm{id} + \lambda^{k+1} Z_{k+1} + \dots. \tag{3.7}$$

The LHS is an Poisson automorphism what, at zeroth order, is identity. This means one can infer that $\exists Z_{(k+1)} \in \lambda^{k+1}\,\mathrm{D}(\mathscr{B})[[\lambda]]$ such that the RHS is its formal power series. That is

$$\exp(Z_{(k+1)}) = \mathrm{id} + \lambda^{k+1} Z_{k+1} + \dots. \tag{3.8}$$

Equation (3.8), as we already stated, defines the dissonance between the Poisson morphisms $\exp(\overline{X})$ and $\exp(Y_{(k)})\exp(X)\exp(V_{(k)})$ at order $k+1$. Consider the vertical and Poisson derivations $V_{k=1}, Y_{k+1} \in \lambda\,\mathrm{D}(\mathscr{B})[[\lambda]]$ and, for simplicity, define $\Gamma_X(Y_{(k)}, V_{(k)}) := \exp(Y_{(k)})\exp(X)\exp(V_{(k)})$. We compute the equivalent of (3.7), where we define the second error term, for the order $k+1$:

$$\begin{aligned}\exp(\lambda^{k+1}Y_{k+1})\Gamma_X(Y_{(k)}, V_{(k)})\exp(\lambda^{k+1}V_{k+1})\exp(-\overline{X}) &= (\mathrm{id} + \lambda^{k+1}Y^0_{k+1} + \dots)\Gamma_X(Y_{(k)}, V_{(k)})(\mathrm{id} + \\ &\quad + \lambda^{k+1}V^0_{k+1} + \dots) \\ &= \mathrm{id} + \lambda^{k+1}Z_{k+1} + \lambda^{k+1}\big(Y^0_{k+1} + V^0_{k+1}\big) + \dots,\end{aligned} \tag{3.9}$$

Again, because the error of order $k+1$ must be zero, we have that

$$0 = \lambda^{k+1}\big(Z_{k+1} + Y^0_{k+1} + V^0_{k+1}\big) \tag{3.10}$$

meaning,

$$-Z_{k+1} = Y^0_{k+1} + V^0_{k+1}. \tag{3.11}$$

This defines a compatibility condition. It states that the existing error Z_{k+1} in order $k+1$ can be canceled exactly if it can be decomposed as $Y^0_{k+1} + V^0_{k+1}$, where Y^0_{k+1} is a Poisson derivation and V^0_{k+1} is vertical. We then compute by substitution $\exp(Z_{(k+1)})\sigma(\exp(-Z_{(k+1)})\Gamma_X(Y_{(k)}, V_{(k)})\phi_0(a_1), a_1 \leftarrow a_2)$:

$$\begin{aligned} \text{RHS} &= \Gamma_X(Y_{(k)}, V_{(k)}) \exp(-\overline{X})\sigma(\exp(\overline{X})\phi_0(a_1), a_1 \leftarrow a_2) \\ &= \sigma(\Gamma_X(Y_{(k)}, V_{(k)})\phi_0(a_1), \Gamma_X(Y_{(k)}, V_{(k)})\phi_0(a_2)) \end{aligned} \tag{3.12}$$

And then, instead of using substitutions, we do the same computation once again, but by expanding the RHS:

$$\begin{aligned} \text{RHS} &= (\mathrm{id} + \lambda^{k+1} Z_{k+1} + \dots)\sigma\big((\mathrm{id} - \lambda^{k+1} Z_{k+1} + \dots)\Gamma_X(Y_{(k)}, V_{(k)})\phi_0(a_1), a_1 \leftarrow a_2\big) \\ &= \sigma\big(\Gamma_X(Y_{(k)}, V_{(k)})\phi_0(a_1), a_1, \leftarrow a_2\big) + \lambda^{k+1}\big(Z_{k+1}\phi_0\{a_1, a_2\}_{\mathscr{A}} - \{Z_{k+1}\phi_0(a_1), \phi_0(a_2)\}_{\mathscr{B}} - \\ &- \{\phi_0(a_1), \Gamma_X(Y_{(k)}, V_{(k)})\phi_0(a_2)\} + \dots \\ &= \sigma\big(\Gamma_X(Y_{(k)}, V_{(k)})\phi_0(a_1), \Gamma_X(Y_{(k)}, V_{(k)})\phi_0(a_2)\big) - \lambda^{k+1}(\delta\phi_0^* Z_{k+1})(a_1, a_2) + \dots \end{aligned} \tag{3.13}$$

For the two solutions X and $\overline{X}$ to be equivalent, the error at order $k+1$ must vanish. Therefore, from (3.13) we get the cocycle condition

$$\delta\phi_0^* Z_{k+1} = 0 \tag{3.14}$$

which implies that $\phi_0^* Z_{k+1}$ is a 1-cocycle in $C^\bullet_{CE,der}(\mathscr{A}, \mathscr{B})$. Since, by hypothesis, $H^1_{CE,der}(\mathscr{A}, \mathscr{B}) = \{0\}$, every 1-cocycle in an 1-coboundary which, by definition, means that $\exists H_{k+1} \in C^0_{CE,der}(\mathscr{A}, \mathscr{B})$ 0-cochain such that $\phi_0^* Z_{k+1} = \delta H_{k+1}$. It follows from the definition of δ that

$$(\delta H_{k+1})(a) = -\{H_{k+1}, \phi_0(a)\}_{\mathscr{B}} \Rightarrow Z_{k+1} = -\{H_{k+1}, \phi_0(a)\}_{\mathscr{B}} \tag{3.15}$$

From (3.11), we have: $V^0_{k+1} = \{H_{k+1}, \cdot\}_{\mathscr{B}} - Y^0_{k+1}$. V^0_{k+1} is, by hypothesis, a vertical derivation. We now only need to choose a proper Y^0_{k+1} to preserve this property. A natural choice is $Y^0_{k+1} = \{H_{k+1}, \cdot\}_{\mathscr{B}}$. Let's verify:

$$V^0_{k+1}\phi_0(a) = \{H_{k+1}, \phi_0(a)\}_{\mathscr{B}} - \{H_{k+1}, \phi_0(a)\}_{\mathscr{B}} = 0 \tag{3.16}$$

Hence, by choosing $V_{k+1} = V^0_{k+1}$, $Y_{k+1} = \sigma(H_{k+1}, \cdot)$ we can cancel the error at order $k+1$. Therefore, $\exp(\overline{X})$ and $\exp(Y_{(k+1)})\exp(X)\exp(V_{(k+1)})$ are equal up to order $k+1$. □

II.. Commutants and Uniqueness

Given a Poisson morphism $\phi_0 : (\mathscr{A}, \pi_0) \to (\mathscr{B}, \sigma_0)$, its image has a Poisson commutant

$$\mathscr{A}' := (\phi_0(\mathscr{A}))^c = \{b \in \mathscr{B} \mid \{b, \phi_0(a)\}_{\mathscr{B}} = 0 \text{ for all } a \in \mathscr{A}\}.$$

This is a Poisson subalgebra of $\mathscr{B}$, inheriting a Poisson structure π_0'. We then have a diagram of Poisson morphisms with Poisson commuting images:

$$(\mathscr{A}, \pi_0) \xrightarrow{\phi_0} (\mathscr{B}, \sigma_0) \xleftarrow{\phi_0'} (\mathscr{A}', \pi_0'),$$

where $\phi_0' : \mathscr{A}' \hookrightarrow \mathscr{B}$ is the inclusion map. Given a formal deformation $\Phi : (\mathscr{A}[[\lambda]], \pi) \to (\mathscr{B}[[\lambda]], \sigma)$ of ϕ_0, we wish to deform the right leg of this diagram.

Proposition 15. *Let $X' \in \lambda\,\mathrm{Der}(\mathscr{B})[[\lambda]]$ be such that $\exp(X')\phi'(\mathscr{A}[[\lambda]]) \subseteq C$. Then*

- *the map $\exp(X')\phi_0' : \mathscr{A}[[\lambda]] \to C$ is a ring homomorphism;*
- *there is a unique Poisson structure π' on $\mathscr{A}[[\lambda]]$ uch that $\Phi' = \exp(X')\phi_0' : (\mathscr{A}'[[\lambda]], \pi') \to (\mathscr{B}[[\lambda]], \sigma)$ is a Poisson morphism.*

W The proof of this proposition can be found in ([1]). Assuming the existence of a derivation $X' \in \lambda\,\mathrm{Der}(\mathscr{B})[[\lambda]]$ such that, for $\Phi' = \exp(X')\phi_0' : \mathscr{A}'[[\lambda]] \to \mathscr{B}[[\lambda]]$, the image $\Phi'(\mathscr{A}'[[\lambda]])$ lies in the commutant of $\Phi(\mathscr{A}[[\lambda]])$ with respect to σ, we then have that it induces a Poisson structure π' on $\mathscr{A}'[[\lambda]]$ deforming π_0', such that

$$(\mathscr{A}[[\lambda]], \pi) \xrightarrow{\Phi} (\mathscr{B}[[\lambda]], \sigma) \xleftarrow{\Phi'} (\mathscr{A}'[[\lambda]], \pi')$$

is a diagram of Poisson morphisms with commuting images. The existence of X' is governed by the vanishing of $H^1_{\mathrm{CE,der}}(\mathscr{A}, \mathscr{B})$, together with the existence of a horizontal lift along ϕ_0'. More precisely, one requires the following two conditions:

1. There exists an $\mathscr{A}'$-linear map $\phi_1 : C^1_{\mathrm{CE,der}}(\mathscr{A},\mathscr{B}) \to C^0_{\mathrm{CE,der}}(\mathscr{A},\mathscr{B})$ such that $\delta D = 0$ implies $D = \delta\phi_1(D)$.
2. There exists a horizontal lift $C^1_{\mathrm{CE,der}}(\mathscr{A}',\mathscr{B}) \to \mathrm{Der}(\mathscr{B})$ along ϕ_0'.

Proposition 16 (Existence of commutant deformations)**.** *Under the above two assumptions, there exists a derivation $X' \in \lambda\,\mathrm{Der}(\mathscr{B})[[\lambda]]$ such that $\Phi'(\mathscr{A}'[[\lambda]])$ lies in the commutant of $\Phi(\mathscr{A}[[\lambda]])$.*

Proof. The proof is, again, by induction on k. For the base case, we take $X'_{(0)} = 0$. So

$$\sigma(\exp(X'_{(0)})\phi_0'(a'), \Phi(a)) = \sigma(\phi_0'(a'), \phi_0(a)) = 0. \tag{3.17}$$

Now, suppose, for the induction hyphotesis, that there exists some $X'_{(k)}$ such that

$$\sigma(\exp(X'_{(k)})\phi_0'(a'), \Phi(a)) = \lambda^{k+1}F_{k+1}(a',a) + \dots \tag{3.18}$$

with $F_{k+1} : \mathscr{A}' \times \mathscr{A} \to \mathscr{A}$ a 1-cochain and a 1-cocycle. By assumption (1), F_{k+1} is also a 1-coboundary. Consider $X'_{k+1} \in \mathrm{Der}(\mathscr{B})$ and define $\exp(\lambda^{k+1}X'_{k+1})\exp(X'_{(k)}) = \Phi'_{(k+1)}$. Then,

$$\begin{aligned}\sigma(\Phi'_{(k+1)}\phi_0'(a'), \Phi(a)) &= \sigma(\lambda^{k+1}X'_{k+1})\exp(X'_{(k)})\phi_0'(a'), \Phi(a)) \\ &= \sigma\big((\mathrm{id} + \lambda^{k+1}X'_{k+1} + \dots)\exp(X'_{(k)})\phi_0'(a'), \Phi(a)\big) \\ &= \lambda^{k+1}F_{k+1}(a',a) + \lambda^{k+1}\{X'_{k+1}\phi_0'(a'), \phi_0(a)\}_{\mathscr{B}} + \dots\end{aligned} \tag{3.19}$$

Again, the error in the order $k+1$ must be zero, so

$$F_{k+1}(a',a) = \{\phi_0(a), X'_{k+1}\phi_0'(a')\}_{\mathscr{B}} \Leftrightarrow \delta X'_{k+1}\phi_0'(a') = F_{k+1}(a',\cdot). \tag{3.20}$$

This is an obstruction condition. Since F_{k+1} inherits the Leibniz rule from σ,

$$F_{k+1}(a'_1a'_2,\cdot) = \phi_0'(a'_1)F_{k+1}(a'_2,\cdot) + \phi_0'(a'_2)F_{k+1}(a'_1,\cdot) \tag{3.21}$$

Finding a suitable X'_{k+1} that satisfies such obstruction is what solves problem (2) up to order $k+1$.

Define $Y(a') := \varphi_1(F_{k+1}(a',\cdot)) \in \mathscr{B}$. One checks $Y \in C^1_{\mathrm{CE,der}}(\mathscr{A}',\mathscr{B})$. By assumption (2), there exists $X'_{k+1} \in \mathrm{Der}(\mathscr{B})$ with $\phi_0'^*(X'_{k+1}) = Y$, i.e.

$$X'_{k+1}(\phi_0'(a')) = Y(a'). \tag{3.22}$$

Then substituting gives the required equation (3.20). This X'_{k+1} extends the deformation to order $k+1$. □

Once a solution to the commutant problem is found, it defines a formal Poisson structure π' on $\mathscr{A}'[[\lambda]]$ uniquely up to equivalence.

Proposition 17. *Let X' and X'' be two solutions to the commutant problem. Then there exists a derivation $\xi \in \lambda\,\mathrm{Der}(\mathscr{A}')[[\lambda]]$ such that*

$$\exp(X')\phi_0' = (\exp(X'')\phi_0') \circ \exp(\xi).$$

In particular, the induced Poisson structures π' and π'' on $\mathscr{A}'[[\lambda]]$ are equivalent.

Proof. For each $a' \in \mathscr{A}'[[\lambda]]$, there is a unique element $a'' \in \mathscr{A}'[[\lambda]]$ such that

$$\exp(X')\phi_0'(a') = \exp(X'')\phi_0'(a''),$$

since both maps are isomorphisms onto the commutant C of $\Phi(\mathscr{A}[[\lambda]])$ in $\mathscr{B}[[\lambda]]$. The assignment $a' \mapsto a''$ defines an automorphism of $\mathscr{A}'[[\lambda]]$ starting at the identity, hence is of the form $\exp(\xi)$ for some $\xi \in \lambda\,\mathrm{Der}(\mathscr{A}')[[\lambda]]$. This proves the equivalence of the two solutions. Consequently, the induced Poisson structures are equivalent via $\exp(\xi)$. □

Combining the existence and uniqueness results for both the deformation of ϕ_0 and the deformation of the commutant, we obtain a well-defined map

$$\mathrm{FPois}_{\sigma_0}(\mathscr{B}) \times \mathrm{FPois}_{\pi_0}(\mathscr{A}) \to \mathrm{FPois}_{\pi_0'}(\mathscr{A}'), \qquad ([\sigma],[\pi]) \mapsto [\pi'],$$

where $\mathrm{FPois}_{\pi_0}(\mathscr{A})$ denotes the set of equivalence classes of formal Poisson structures on $\mathscr{A}[[\lambda]]$ deforming π_0. This map is independent of all choices involved in the construction, as a consequence of the uniqueness results.

Proposition 18. *Suppose we have two diagrams of Poisson morphisms with commuting images deforming the same underlying diagram,*

$$(\mathscr{A}[[\lambda]], \pi^{(i)}) \xrightarrow{\Phi^{(i)}} (\mathscr{B}[[\lambda]], \sigma^{(i)}) \xleftarrow{\Phi^{(i)'}} (\mathscr{A}'[[\lambda]], \pi^{(i)'}), \qquad i = 1, 2,$$

and suppose there are Poisson isomorphisms

$$\psi : (\mathscr{A}[[\lambda]], \pi^{(1)}) \to (\mathscr{A}[[\lambda]], \pi^{(2)}), \qquad \Psi : (\mathscr{B}[[\lambda]], \sigma^{(1)}) \to (\mathscr{B}[[\lambda]], \sigma^{(2)})$$

such that $\Phi^{(2)} \circ \psi = \Psi \circ \Phi^{(1)}$. *Then there exists a Poisson isomorphism*

$$\psi' : (\mathscr{A}'[[\lambda]], \pi^{(1)'}) \to (\mathscr{A}'[[\lambda]], \pi^{(2)'})$$

such that $\Phi^{(2)'} \circ \psi' = \Psi \circ \Phi^{(1)'}$.

This proposition ensures that the map $([\sigma], [\pi]) \mapsto [\pi']$ is well-defined on equivalence classes, providing a robust algebraic framework for studying deformations of bimodules.

Remark 10. *In the geometric setting of interest, where* $\mathscr{A} = C^\infty(P)$ *with the zero Poisson structure and* $\mathscr{B} = C^\infty(T^*P)$ *with a symplectic Poisson structure coming from* $\omega_{\text{can}} + \rho^* B_0$, *the cohomological conditions required above are satisfied. The relevant cohomology groups vanish due to a chain homotopy constructed using the vertical de Rham differential, and horizontal lifts exist via the choice of a connection on the cotangent bundle. This allows us to deform the geometric equivalence bimodules for trivial Poisson structures in a completely unobstructed manner.*

III.. The Classifying Action and its Identification with Gauge Transformations

1.. Geometric Setup and Cohomological Vanishing

For the entirety of this subsection, suppose that $P_1 = P_2 = P$ and consider the diagram

$$(C^\infty(P)[[\lambda]], \pi^{(1)}) \xrightarrow{\Phi(1)} (C^\infty(S)[[\lambda]], \omega) \xleftarrow{\Phi(2)} (C^\infty(P)[[\lambda]], \pi^{(2)}), \tag{3.23}$$

with $\pi_0^{(1)} = \pi_0^{(2)} = 0$. One can make the assumption that $P_1 = P_2$ due to the fact that, when the Poisson structure vanishes at zeroth order, one have, for $\lambda = 0$, the following diagram

$$(P_1, 0) \xleftarrow{J_1} (S, \omega_0) \xrightarrow{J_2} (P_2, 0), \tag{3.24}$$

meaning P_1 and P_2 must be diffeomorphic. Moreover, (3.24) is necessarily isomorphic to the bimodule

$$(P, 0) \xleftarrow{\rho} (T^*P, \omega_0) \xrightarrow{J_2} (P, 0) \tag{3.25}$$

with $\rho : T^*P \to P$ the natural projection and $\omega_0 := \omega_{B_0} = \omega_{\text{can}} + \rho^* B_0$, given that B_0 is a closed 2-form on P. Using this result, one have that the Poisson morphism that will proceed the formal deformation process is $\rho^* : \mathscr{A} \to \mathscr{B}$, where we define $\mathscr{A} := C^\infty(P)$ and $\mathscr{B} := C^\infty(T^*P)$, that at zeroth order is simply $\rho : (T^*P, \omega_{B_0})) \to (P, 0)$. As expected, and given the definitions of $\mathscr{A}$ and $\mathscr{B}$, the associated Chevalley-Eilenberg complex is totally analogous to what we had in section (**??**), meaning that $C^k_{CE,der}(P, T^*P) = \Gamma(\rho^* \wedge^k T^*P)$, with elements $D : C^\infty(P) \times \cdots \times C^\infty(P) \to C^\infty(T^*P)$ that are not only skew-symmetric and k-linear maps, but also derivations, and whose differentials are given by

$$(\delta D)(f_0, \dots, f_k) = \sum_{j=0}^{k} (-1)^j \{\rho^*(f_j), D(f_0, \dots, \hat{f_j}, \dots, f_k)\}_{\text{can}}. \tag{3.26}$$

One of the main results of this section, is tha vanishing of the cohomology associated to $C^\bullet_{CE,der}(P, T^*P)$. To prove this result, one needs the following lemma, that will be stated but not proved:

Lemma III..1. *The differential δ is $C^\infty(P)$-linear*

Its complete proof can be found in details on ([1]). Therefore:

Proposition 19. *For $k \geq 1$, there exists a sequence h of $C^\infty(P)$-linear maps $h_k : C^k_{CE,der}(P, T^*P) \to C^{k-1}_{CE,der}(P, T^*P)$ satisfying $\delta h + h\delta = \text{id}$. In particular, $H^k_{CE,der}(P, T^*P) = \{0\}$ for $k \geq 1$.*

Proof. The ideia of the proof is to prove it locally, than extend it globally. For the first step, consider the usual de Rham homotopy operator on $\mathbb{R}^n$, $h_{dR} : \Omega^\bullet(\mathbb{R}^n) \to \Omega^{\bullet-1}(\mathbb{R}^n)$, which satisfies $dh_{dR} + h_{dR}d = \text{id}$. Next, take a local chart $U \subseteq P$ so that $T^*P = U \times \mathbb{R}^n$. Let $E \cong U \times \mathbb{R}^n$ with local coordinates $(q^1, \dots, q^n)$, where $\pi : E \to U$ is the projection onto the q-coordinates. A vertical differential form is a section of $\wedge^\bullet \ker(d\pi)^*$, and in local coordinates it has the form

$$\omega(q, p) = \sum_I a_I(q, p) dp_I. \tag{3.27}$$

For each fixed base point q, the restriction $\omega|_q := \omega|_{\pi^{-1}(q)}$ is an ordinary differential form on the fiber $\pi^{-1}(q) \cong \mathbb{R}^n = \{(p_1, \dots, p_n)\}$. Thus, one may regard a vertical form as a smooth familiy of forms on $\mathbb{R}^n$. The de Rham homotopy operator is then applied fiberwise by defining $(h_{dR}\omega))_q := h_{dR}(\omega_q)$, or equivalently, $(d_{dR}\omega)(q, p) = h_{dR}(\omega(q, \cdot))(p)$, where, in this case, q is treated as a parameter and $h_{dR} : \Omega^\bullet(\mathbb{R}^n) \to \Omega^{\bullet-1}(\mathbb{R}^n)$ acts only on p. This induces a map $h_{ver} : \Omega^\bullet_V(\mathbb{R}^n) \to \Omega^{\bullet-1}_V(\mathbb{R}^n)$ that is booth $C^\infty(U)$-linear and satisfies $d_{ver}h_{ver} + h_{ver}d_{ver} = \text{id}$. By lemma (III..1), there exists a isomorphism of complexes $\Psi : (C^\bullet_{CE,der}(U, T^*U), \delta) \to (C^{\bullet-1}_{CE,der}(U, T^*U), \delta)$ that is used to turn h_{ver} into a $C^\infty(U)$-liear homotopy operator $h : C^\bullet_{CE,der}(U, T^*U) \to C^{\bullet-1}_{CE,der}(U, T^*U)$, proving that the results holds locally.

To extend it globally, consider the atlas $\{U_\alpha\}$ of P, and let $\{\varphi_\alpha\}$ be a locally finite partition of unity subordinate to it, that is, $\text{supp}(\varphi_\alpha) \subset U_\alpha$ and

$$\sum_\alpha \varphi_\alpha(x) = 1, \tag{3.28}$$

for every $x \in P$. For each α, we have the $C^\infty(U)$-linear homotopy operator

$$h_\alpha : C^\bullet_{CE,der}(U_\alpha, T^*U_\alpha) \to C^{\bullet-1}_{CE,der}(U_\alpha, T^*U_\alpha) \tag{3.29}$$

that, of course, satisfies $\delta_\alpha h_\alpha + h_\alpha \delta_\alpha = \text{id}$, with δ_α a restriction (in the usual sense) of δ. Now, define

$$h := \sum_\alpha \varphi_\alpha h_\alpha, \tag{3.30}$$

and notice that using the $C\infty(U_\alpha)$-linearity of δ_α, one have that $\varphi_\alpha \,\mathrm{id} = \varphi_\alpha(\delta_\alpha h_\alpha + h_\alpha\delta_\alpha) = \delta_\alpha(\varphi_\alpha h_\alpha) + (\varphi_\alpha h_\alpha)\delta_\alpha$, meaning that h is $C^\infty(U)$-linear and that

$$\delta h + h\delta = \sum_\alpha \delta_\alpha(\varphi_\alpha h_\alpha) + (\varphi_\alpha h_\alpha)\delta_\alpha = \sum_\alpha \mathrm{id} = \mathrm{id}, \tag{3.31}$$

which completes the proof. □

2.. Formal Deformations and the Classifying Action

With the main result of the previous subsection, in consonance with the results of section (**??**), we can finally proceed to the main result of this subsection:

Theorem 7. *Given any formal symplectic structure $\omega = \omega_0 + \sum_{k=1}^\infty \lambda^k\omega_k$ on T^*P with $\omega_0 = \omega_{B_0}$, and formal Poisson structure π on P with $\pi_0 = 0$, there exist formal vector fields $X, X' \in \lambda\mathfrak{X}(T^*P)[[\lambda]]$ and formal Poisson structure π' on P, with $\pi'_0 = 0$, such that*

$$(C^\infty(P)[[\lambda]], \pi) \xrightarrow{\Phi} (C^\infty(T^*P[[\lambda]], \omega)) \xleftarrow{\Phi'} (C^\infty(P)[[\lambda]], \pi') \tag{3.32}$$

is an equivalence bimodule, where $\Phi = \exp(\mathscr{L}_X)\rho^*$ and $\Phi' = \exp(\mathscr{L}_{X'})\rho^*$.

Proof. Note that the pullback bundle $\rho^*TP \to T^*P$, induced by the cotangent bundle projection $\rho : T^*P \to P$, is the vector bundle whose fiber over a point $\xi \in T^*_xP$ is $(\rho^*TP) * \xi = T_xP$. Thus, a section $X \in \Gamma(\rho^*TP)$ is not necessarily a vector field on P; rather, it assigns to each covector $\xi \in T^*_xP$ a vector $X(\xi) \in T_xP$, depending smoothly on ξ. Now suppose that we have chosen a horizontal distribution $H \subset T(T^*P)$ such that $T(T^*P) = H \oplus \ker(d\rho)$. Since $d\rho : H * \xi \to T_xP$ is an isomorphism for every $\xi \in T^*_xP$, each vector $v \in T_xP$ admits a unique horizontal lift $v^H_\xi \in H_\xi$, in the sense defined in Section **??**, satisfying $d\rho(v^H_\xi) = v$.

Now let $X \in \Gamma(\rho^*TP)$. For each $\xi \in T^*P$, the vector $X(\xi)$ belongs to $T_{\rho(\xi)}P$, and therefore there exists a unique horizontal vector $X^H(\xi) \in H_\xi$ such that $d\rho(X^H(\xi)) = X(\xi)$. In this way, one obtains a map $\Gamma(\rho^*TP) \to \mathfrak{X}(T^*P)$ given by $X \mapsto X^H$, where $X^H(\xi) := (X(\xi))^H$. This map is, again, a horizontal lift in the sense of Section **??**. Since $H^2_{CE,der}(P, T * P) = \{0\}$, by Proposition 19, pProposition 13 implies the existence of a formal vector field X such that

$$\Phi = \exp(\mathscr{L}_X)\rho^* : (C^\infty(P)[[\lambda]], \pi) \to (C^\infty(T^*P)[[\lambda]], \omega) \tag{3.33}$$

is a Poisson morphism. Fix X. The Poisson commutator of $\rho^*C^\infty(P)$ in $C^\infty(T^*P)$ agrees with itself, so we can use the results of the previous section directly: condition (2) used in Proposition 16 stablishes the existence of a horizontal lift $\Gamma(\rho^*TP) \to \mathfrak{X}(T^*P)$, which we just showed that exist. Condition (1), on the other hand, holds by Proposition 19. As a consequence, there exist a formal vector field X' such that the image of $\Phi' = \exp(\mathscr{X}')\rho^*$ Poisson commutes with the image of Φ with respect to ω. So, by pProposition 15, the choice of X' uniquely determines a formal Poisson structure π' on P so that $\Phi' : (C^\infty(P)[[\lambda]], \pi') \to (C^\infty(T^*P)[[\lambda]], \omega)$ is Poisson (or anti-Poisson), which completes the proof.

□

Analogously to Section **??**, this construction, together with its uniqueness properties, packages naturally into a map

$$\mathrm{FPois}_{\sigma B_0}(T^*P) \times \mathrm{FPois}_0(P) \longrightarrow \mathrm{FPois}_0(P), \tag{3.34}$$

which we denote by γ_{B_0}, in order to record its dependence upon the initial choice of symplectic form $\omega_{B_0} = \omega_{\mathrm{can}} + \rho^*B_0$ on T^*P. The identification $\mathrm{FPois}_{\sigma B_0}(T^*P) = \lambda H^2_{\mathrm{dR}}(T^*P)[[\lambda]]$ rests upon three observations. Firstly, since $\omega_{B_0} = \omega_{\mathrm{can}} + \rho^*B_0$ is symplectic, it is formally invertible in the sense of Section 2.1 of [1], so that formal Poisson structures deforming $\sigma_{B_0} = (\omega_{B_0})^{-1}$ correspond bijectively to formal symplectic forms

$$\omega = \omega_{B_0} + \sum_{k=1}^\infty \lambda^k\omega_k, \qquad \omega_k \in \Omega^2(T^*P) \text{ closed.} \tag{3.35}$$

Secondly, two such formal Poisson structures $\pi^{(1)}$ and $\pi^{(2)}$, corresponding to formal symplectic forms $\omega^{(1)}$ and $\omega^{(2)}$, are equivalent — that is, related by a formal diffeomorphism $\exp(\mathscr{L}X)$ with $X \in \lambda\mathfrak{X}(T^*P)[[\lambda]]$ — precisely when $\omega^{(1)}$ and $\omega^{(2)}$ are cohomologous order by order in λ, i.e. when $[\omega^{(1)}_k] = [\omega^{(2)}_k] \in H^2_{\mathrm{dR}}(T^*P)$ for every $k \geqslant 1$; this is the formal counterpart of the classical fact that $\exp(\mathscr{L}_X)$ preserves de Rham cohomology classes. Thirdly, it follows that the set of equivalence classes is parametrised precisely by a sequence of cohomology classes $([\omega_1], [\omega_2], \ldots)$, that is, by an element of

$$\lambda H^2_{\mathrm{dR}}(T^*P)[[\lambda]] = \left\{ \sum_{k=1}^\infty \lambda^k c_k \;:\; c_k \in H^2_{\mathrm{dR}}(T^*P) \right\}. \tag{3.36}$$

Consequently, $\mathrm{FPois}_{\sigma B_0}(T^*P)$ is an affine space modelled upon the abelian group $\lambda H^2_{\mathrm{dR}}(T^*P)[[\lambda]]$, with base point σ_{B_0} corresponding to the zero series; the group acts by translation, and it is in this sense that one identifies $\mathrm{FPois}_{\sigma B_0}(T^*P)$ with $\lambda H^2_{\mathrm{dR}}(T^*P)[[\lambda]]$ itself.

The second detail concerns the isomorphism $H^2_{\mathrm{dR}}(P) \cong H^2_{\mathrm{dR}}(T^*P)$, which is given explicitly by pullback along the bundle projection $\rho : T^*P \to P$, namely

$$\rho^* : H^2_{\mathrm{dR}}(P) \longrightarrow H^2_{\mathrm{dR}}(T^*P), \qquad [B] \longmapsto [\rho^* B]. \tag{3.37}$$

This map is an isomorphism because ρ is a vector bundle, and every vector bundle deformation retracts onto its zero section, via the fibrewise contraction $\xi \mapsto t\xi$, $t \in [0,1]$, taken down to $t = 0$; the projection ρ is thus a homotopy equivalence, and pullback in de Rham cohomology along a homotopy equivalence is an isomorphism, exactly as in the fibred Poincaré lemma underlying the identity $H^\bullet_{\mathrm{dR}}(\mathbb{R}^n) \cong H^\bullet_{\mathrm{dR}}(\{0\})$. Its inverse is induced by restriction along the zero section $s_0 : P \hookrightarrow T^P$, so that $(\rho^*)^{-1} = s_0^*$. With this map made explicit, one may write γ_{B_0} concretely as

$$\gamma_{B_0} : \lambda H^2_{\mathrm{dR}}(P)[[\lambda]] \times \mathrm{FPois0}(P) \longrightarrow \mathrm{FPois}_0(P), \quad \gamma B_0\Big(\Big[\sum_{k=1}^{\infty} \lambda^k B_k\Big], [\pi]\Big) := \gamma_{B_0}\Big(\Big[\sum_{k=1}^{\infty} \lambda^k \rho^* B_k\Big], [\pi]\Big), \tag{3.38}$$

where the right-hand side is read through the identification just described, namely

$$\sum_k \lambda^k \rho_k^B \in \lambda H^2_{\mathrm{dR}}(T^*P)[[\lambda]] = \mathrm{FPois}_{\sigma B_0}(T^*P).$$

Concretely, given representatives $\sum_{k=1}^{\infty} \lambda^k B_k$ and π of classes in $\lambda H^2_{\mathrm{dR}}(P)[[\lambda]]$ and $\mathrm{FPois0}(P)$ respectively, the resulting class in $\mathrm{FPois}_0(P)$ is defined by any formal Poisson structure π' fitting into an equivalence bimodule of the form 3.24, taking $\omega = \omega_{\mathrm{can}} + \rho^* B_0 + \rho^!\left(\sum_{k=1}^{\infty} \lambda^k B_k\right)$.

Although γ_{B_0} appears, from its very definition, to depend upon the choice of representative B_0 and not merely upon its cohomology class $[B_0] \in H^2_{\mathrm{dR}}(P)$, this dependence is in fact illusory, as the following lemma shows:

Lemma III..2. *The map γ_{B_0} only depends on the cohomology class of B_0.*

Proof. If $B_0 - B_0' = d\theta$ for $\theta \in \Omega^1(P)$, then fiber-translation by θ defines a symplectomorphism $\psi_\theta : (T^*P, \omega_{B_0}) \to (T^*P, \omega_{B_0'})$ such that $\rho \circ \psi_\theta = \rho$. So, for any closed $b \in \lambda\Omega^2(P)[[\lambda]]$, we have

$$\psi_\theta^*(\omega_{B_0'} + \rho^* b) = \omega_{B_0} + \rho^* b, \tag{3.39}$$

meaning that if π and π' fit into the equivalence bimodule (3.32), then they also fit into an equivalence bimodule with formal symplectic form $\omega = \omega_{B_0'} + \rho^* b$ on T^*P, showing that $\gamma_{B_0}([b],[\pi]) = \gamma_{B_0'}([b],[\pi]) = [\pi']$. □

It follows at once that $\gamma_{B_0} = \gamma_{B_0'}$ on the relevant arguments. This permits one to combine the zeroth-order term B_0 with the higher-order terms $B_1, B_2, \ldots$ into a single closed formal series $B = B_0 + \sum_{k\geqslant 1} \lambda^k B_k \in \Omega^2(P)[[\lambda]]$, and to define the classifying action

$$\gamma : H^2_{\mathrm{dR}}(P)[[\lambda]] \times \mathrm{FPois0}(P) \longrightarrow \mathrm{FPois}_0(P), \qquad \gamma([B],[\pi]) := \gamma B_0([b],[\pi]), \quad b := \sum_{k=1}^{\infty} \lambda^k B_k. \tag{3.40}$$

By construction, any representative π^B of $\gamma([B],[\pi])$ is Morita equivalent to π, since it arises from an explicit equivalence bimodule of the form (3.23). We are now in a position to state the theorem that this classifying action was built to serve, recalling that the natural action of $\mathrm{Diff}(P)$ upon formal Poisson structures descends to an action upon $\mathrm{FPois}_0(P)$ by $\psi[\pi] = [\psi_\pi]$.

Theorem 8. *Two formal Poisson structures π and π' on P, both vanishing in zeroth order, are Morita equivalent if and only if there exist $[B] \in H^2_{\mathrm{dR}}(P)[[\lambda]]$ and $\psi \in \mathrm{Diff}(P)$ such that $[\pi'] = \psi_* \gamma([B],[\pi])$.*

Sketch. The implication from right to left is immediate, since representatives of $\gamma([B],[\pi])$ are Morita equivalent to π by construction, whilst diffeomorphisms preserve Morita equivalence classes by definition. For the converse, one takes an equivalence bimodule between π and π'; its underlying zeroth-order diagram is a self-Morita-equivalence of $(P,0)$, and hence, by the classical result, must be of the form $S = T^*P$, $J_1 = \rho$, $J_2 = \psi \circ \rho$, $\omega_0 = \omega_{\mathrm{can}} + \rho^* B_0$, for some closed $B_0 \in \Omega^2(P)$ and some $\psi \in \mathrm{Diff}(P)$. Each higher-order term ω_k, $k \geqslant 1$, being closed, is cohomologous to $\rho^* B_k$ for some closed $B_k \in \Omega^2(P)$, so that a further formal vector field identifies ω with $\omega_{\mathrm{can}} + \rho^* B$, where $B = B_0 + \sum_{k\geqslant 1} \lambda^k B_k$; precomposing the legs of the resulting bimodule then produces precisely the standard form (60), from which one reads off that $[\psi^*\pi'] = \gamma([B],[\pi])$, that is, $[\pi'] = \psi_*\gamma([B],[\pi])$, as required. □

3.. Self-Equivalence Bimodules via Dirac Geometry and Identification with Gauge Transformations

Lemma III..3. *Any formal Poisson structure π on P that vanishes at zeroth order, admits a self equivalence bimodule of the following type:*

$$(C^\infty(P)[[\lambda]], \pi) \xrightarrow{\Phi} (C^\infty(T^*P)[[\lambda]], \omega) \xleftarrow{\Phi'} (C^\infty(P)[[\lambda]], \pi), \tag{3.41}$$

where $\omega = \omega_{can} + \sum_{k=1}^{\infty} \lambda^k d\theta_k$, $\Phi = \rho^*$ and $\Phi' = \exp(\mathscr{L}_X \rho^*)$.

Before we prove Lemma III..3, we need some contextualization on Dirac Geometry. Recall that for a smooth manifold M the *generalized tangent bundle* is $\mathbb{T}M := TM \oplus T^*M$, with space of smooth sections

$$\Gamma(\mathbb{T}M) = \mathfrak{X}(M) \oplus \Omega^1(M).$$

This bundle carries two basic structures making it into the (standard) *Courant algebroid* over M:

- a symmetric $C^\infty(M)$-bilinear pairing
$$\langle X \oplus \xi,\, Y \oplus \eta \rangle := \iota_X \eta + \iota_Y \xi,$$
- We adopt the Courant-Dorfman bracket convention stated earlier in Section 2, namely,
$$[[X \oplus \alpha,\, Y \oplus \beta]] := [X, Y] \oplus \big(\mathscr{L}_X \beta - \iota_Y d\alpha\big),$$

together with the anchor map given by the projection $\mathbb{T}M \to TM$.

A (formal) *Poisson structure* $\pi = \sum_{k\geq 0} \lambda^k \pi_k$ on M can be encoded Dirac-theoretically by its *graph*

$$\mathrm{gr}(\pi) := \big\{ \pi^\sharp(\alpha) \oplus \alpha \;\big|\; \alpha \in \Omega^1(M)[[\lambda]] \big\} \subseteq \Gamma(\mathbb{T}M)[[\lambda]].$$

This is a Lagrangian $C^\infty(M)[[\lambda]]$-submodule with respect to $\langle \cdot, \cdot \rangle$ (i.e. $\langle \cdot, \cdot \rangle$ vanishes identically on it, and it has maximal rank), and the Jacobi/integrability identity $[\pi, \pi] = 0$ for π is *equivalent* to the involutivity of $\mathrm{gr}(\pi)$ under $[[\cdot, \cdot]]$:

$$[[\, \mathrm{gr}(\pi),\, \mathrm{gr}(\pi)\,]] \subseteq \mathrm{gr}(\pi).$$

All the algebraic structures above extend $\mathbb{K}[[\lambda]]$-bilinearly to $\Gamma(\mathbb{T}M)[[\lambda]]$ without any changes to the formulas, and we use them freely below.

Symmetries. A *symmetry* of $\Gamma(\mathbb{T}M)[[\lambda]]$ is a pair (F, ϕ), where $\phi : C^\infty(M)[[\lambda]] \to C^\infty(M)[[\lambda]]$ is an algebra automorphism and $F : \Gamma(\mathbb{T}M)[[\lambda]] \to \Gamma(\mathbb{T}M)[[\lambda]]$ is $\mathbb{K}[[\lambda]]$-linear, satisfying

$$F(ge) = \phi^{-1}(g) F(e), \qquad \phi\langle F(e_1), F(e_2)\rangle = \langle e_1, e_2 \rangle, \qquad [[F(e_1), F(e_2)]] = F([[e_1, e_2]]),$$

for all $e, e_1, e_2 \in \Gamma(\mathbb{T}M)[[\lambda]]$ and $g \in C^\infty(M)[[\lambda]]$. Two basic families of symmetries will be used:

- *Formal diffeomorphisms.* For $X \in \lambda\mathfrak{X}(M)[[\lambda]]$,
$$\exp(-\mathscr{L}_X)(Y \oplus \beta) := \exp(-\mathscr{L}_X) Y \oplus \exp(-\mathscr{L}_X)\beta, \qquad \phi = \exp(\mathscr{L}_X).$$
- *B-field (gauge) transformations.* For a closed $B \in \Omega^2(M)[[\lambda]]$,
$$\tau_B(Y \oplus \beta) := Y \oplus (\beta + \iota_Y B), \qquad \phi = \mathrm{id}.$$

The relevance of τ_B for Poisson geometry is that it implements gauge transformations of Poisson structures precisely as

$$\tau_B(\mathrm{gr}(\pi)) = \mathrm{gr}(\tau_B(\pi)).$$

Infinitesimal symmetries. A *derivation* of $\Gamma(\mathbb{T}M)[[\lambda]]$ is a pair (D, X) with $X \in \mathfrak{X}(M)[[\lambda]]$ and $D : \Gamma(\mathbb{T}M)[[\lambda]] \to \Gamma(\mathbb{T}M)[[\lambda]]$ a $\mathbb{K}[[\lambda]]$-linear map satisfying the corresponding Leibniz-type identities relative to $\mathscr{L}_X$. Every pair $(X, b) \in \lambda\mathfrak{X}(M)[[\lambda]] \oplus \Omega^2(M)[[\lambda]]$ with $db = 0$ determines a derivation via

$$(X, b)(Y \oplus \beta) = [X, Y] \oplus \big(\mathscr{L}_X \beta - \iota_Y b\big), \tag{$*$}$$

and $(*)$ generates the 1-parameter group of symmetries

$$F_t = \exp(-t\mathscr{L}_X)\, \tau_{B_t}, \qquad B_t = \int_0^t \exp(s\mathscr{L}_X) b\, ds. \tag{$**$}$$

A key structural fact, used below, is that $\mathrm{gr}(\pi)$ is involutive under $[[\cdot, \cdot]]$, so *any* derivation that preserves $\mathrm{gr}(\pi)$ at the infinitesimal level preserves it under the full flow $(**)$.

Backward images. For a surjective submersion $J : S \to P$ and a formal Poisson structure π on P, the *backward image* $J^!\,\mathrm{gr}(\pi) \subseteq \Gamma(\mathbb{T}S)[[\lambda]]$ is the $C^\infty(S)[[\lambda]]$-submodule of elements $X \oplus \alpha$ such that, at every $z \in S$ with $q = J(z)$, there is $\xi \in T^*_q P[[\lambda]]$ with

$$(T_z J)^* \xi = \alpha_z, \qquad T_z J(X_z) = \pi_q^\sharp(\xi).$$

This is the formal analogue of the backward image of a Dirac structure along a smooth map, and it is again involutive for $[[\cdot,\cdot]]$. Finally, we can prove the Lemma, that will follow from a more precise statement, Proposition 20, which produces an explicit self-equivalence bimodule using a choice of linear connection on T^*P.

Setup. Fix a linear connection ∇ on the vector bundle $\rho : T^*P \to P$, and let

$$\mathrm{hor} : \rho^*(TP) \longrightarrow \mathfrak{X}(T^*P)$$

be the associated horizontal lift. Using hor, define a formal vector field $Z \in \lambda\mathfrak{X}(T^*P)[[\lambda]]$ by

$$Z_\xi := \mathrm{hor}_\xi\big(\pi^\sharp(\xi)\big), \qquad \xi \in T^*P.$$

(Since $\pi_0 = 0$, we have $\pi^\sharp \in \lambda\,\mathrm{Hom}(T^*P, TP)[[\lambda]]$, so indeed $Z \in \lambda\mathfrak{X}(T^*P)[[\lambda]]$.)

Furthermore, for the following lemma, consider the diagram

$$(C^\infty(P_1)[[\lambda]], \pi^{(1)}) \xrightarrow{\Phi^{(1)}} (C^\infty(S)[[\lambda]], \omega) \xleftarrow{\Phi^{(2)}} (C^\infty(P_2)[[\lambda]], \pi^{(2)}), \tag{3.42}$$

where $\pi^{(1)}$ and $\pi^{(2)}$ are formal Poisson structres, ω a formal symplectic structure and $\Phi^{(i)} = \exp(Z^{(i)})J_i^*$, given $Z^{(i)} \in \lambda\mathfrak{X}(S)[[\lambda]]$ and $J_i : S \to P$ a surjective submersion, for $i = 1, 2$.

Lemma III..4. *Suppose that in (3.42) we have* $\dim(S) = \dim(P_1) + \dim(P_2)$ *and*

$$\tau_{-\omega}((\Phi^{(1)})^!\,\mathrm{gr}(\pi^{(1)})) = (\Phi^{(2)})^!\,\mathrm{gr}(\pi^{(2)})), \tag{3.43}$$

then it is a bimodule.

The proof for Lemma III..4 and be found on [1], section 5.

Proposition 20. *The diagram*

$$(C^\infty(P)[[\lambda]], \pi) \xrightarrow{\rho^*} (C^\infty(T^*P)[[\lambda]], \omega) \xleftarrow{\exp(\mathscr{L}_Z)\rho^*} (C^\infty(P)[[\lambda]], \pi),$$

where $\omega := \int_0^1 \exp(s\mathscr{L}_Z)\omega_{\mathrm{can}}\,ds$*, is an equivalence bimodule.*

Proof. Let $\theta_{\mathrm{can}} \in \Omega^1(T^*P)$ be the tautological 1-form, so that $\omega_{\mathrm{can}} = -d\theta_{\mathrm{can}}$. Consider the derivation $(*)$ of $\Gamma(\mathbb{T}(T^*P))[[\lambda]]$ associated with the pair $(Z, d\theta_{\mathrm{can}})$, i.e.

$$[[Z \oplus \theta_{\mathrm{can}}, \cdot\,]].$$

By $(**)$, this derivation integrates to the 1-parameter group of symmetries

$$F_t = \exp(-t\mathscr{L}_Z)\,\tau_{B_t}, \qquad B_t = -\int_0^t \exp(s\mathscr{L}_Z)\,\omega_{\mathrm{can}}\,ds.$$

We now verify two pointwise facts at each $\xi \in T^*P$:

$$(\theta_{\mathrm{can}})_\xi = \rho^*\xi, \qquad T_\xi\rho(Z_\xi) = \pi^\sharp(\xi).$$

The first is the defining property of the tautological 1-form; the second follows since hor is a section of $T\rho$, i.e. $T\rho \circ \mathrm{hor} = \mathrm{id}$ (identifying ρ^*TP with the pullback bundle), so

$$T_\xi\rho(Z_\xi) = T_\xi\rho\big(\mathrm{hor}_\xi(\pi^\sharp(\xi))\big) = \pi^\sharp(\xi).$$

Together these two identities say precisely, by the definition of backward image, that

$$Z \oplus \theta_{\mathrm{can}} \;\in\; \rho^!\,\mathrm{gr}(\pi).$$

Since $\rho^!\,\mathrm{gr}(\pi)$ is involutive under $[[\cdot,\cdot]]$ (backward images of Dirac/Poisson graphs along submersions are always involutive), the derivation $[[Z \oplus \theta_{\mathrm{can}}, \cdot\,]]$ preserves $\rho^!\,\mathrm{gr}(\pi)$:

$$[[\, Z \oplus \theta_{\mathrm{can}},\ \rho^!\,\mathrm{gr}(\pi)\,]] \subseteq \rho^!\,\mathrm{gr}(\pi).$$

Consequently its flow preserves $\rho^!\,\mathrm{gr}(\pi)$ as well:

$$F_t\big(\rho^!\,\mathrm{gr}(\pi)\big) = \exp(-t\mathscr{L}_Z)\,\tau_{B_t}\big(\rho^!\,\mathrm{gr}(\pi)\big) = \rho^!\,\mathrm{gr}(\pi) \qquad \text{for all } t.$$

Rearranging (using that $\exp(-t\mathscr{L}_Z)$ is invertible with inverse $\exp(t\mathscr{L}_Z)$) gives

$$\tau_{B_t}\big(\rho^!\,\mathrm{gr}(\pi)\big) = \exp(t\mathscr{L}_Z)\,\rho^!\,\mathrm{gr}(\pi).$$

Setting $t = 1$ and writing $\omega := -B_1 = \int_0^1 \exp(s\mathscr{L}_Z)\omega_{\mathrm{can}}\, ds$, this reads

$$\tau_{-\omega}\big(\rho^! \operatorname{gr}(\pi)\big) = \exp(\mathscr{L}_Z)\, \rho^! \operatorname{gr}(\pi). \tag{$\dagger$}$$

It remains to translate ($\dagger$) into the bimodule condition of Lemma III..4, whose hypothesis is

$$\tau_{-\omega}\big((\Phi^{(1)})^! \operatorname{gr}(\pi^{(1)})\big) = (\Phi^{(2)})^! \operatorname{gr}(\pi^{(2)}),$$

for a diagram $\Phi^{(1)} = J_1^*$, $\Phi^{(2)} = \exp(\mathscr{L}_{Z^{(2)}})J_2^*$. Taking $P_1 = P_2 = P$, $J_1 = J_2 = \rho$, $\pi^{(1)} = \pi^{(2)} = \pi$, $Z^{(2)} = Z$, this is exactly ($\dagger$). Moreover $\dim(T^*P) = 2\dim(P) = \dim(P) + \dim(P)$, so the dimension hypothesis of Lemma III..4 also holds. Applying Lemma III..4, we conclude that

$$(C^\infty(P)[[\lambda]], \pi) \xrightarrow{\ \rho^*\ } (C^\infty(T^*P)[[\lambda]], \omega) \xleftarrow{\ \exp(\mathscr{L}_Z)\rho^*\ } (C^\infty(P)[[\lambda]], \pi)$$

is a bimodule. Since its zeroth order truncation ($\lambda = 0$) is the classical self-equivalence bimodule $(P, 0) \xleftarrow{\rho} (T^*P, \omega_{\mathrm{can}}) \xrightarrow{\rho} (P, 0)$ (as $Z, \omega - \omega_{\mathrm{can}} \in \lambda(\cdots)[[\lambda]]$), it is in fact an *equivalence* bimodule, in the sense of Definition 2.5. □

Proof of Lemma III..3. Apply Proposition 20 with $\Phi = \rho^*$ and $\Phi' = \exp(\mathscr{L}_Z)\rho^*$; this already gives an equivalence bimodule of the required form with $Y := Z$. It remains only to check the stated shape of ω.

By construction,

$$\omega = \int_0^1 \exp(s\mathscr{L}_Z)\, \omega_{\mathrm{can}}\, ds = \int_0^1 \big(\omega_{\mathrm{can}} + s\, \mathscr{L}_Z\omega_{\mathrm{can}} + \tfrac{s^2}{2!}\mathscr{L}_Z^2\omega_{\mathrm{can}} + \cdots\big)\, ds,$$

and since $Z \in \lambda\mathfrak{X}(T^*P)[[\lambda]]$, each term $\mathscr{L}_Z^k\omega_{\mathrm{can}}$ contributes at order $\geq \lambda^k$; thus $\omega - \omega_{\mathrm{can}} \in \lambda\Omega^2(T^*P)[[\lambda]]$. Moreover $\mathscr{L}_Z\omega_{\mathrm{can}} = d(\iota_Z\omega_{\mathrm{can}})$ since ω_{can} is closed, so each term $\frac{s^k}{k!}\mathscr{L}_Z^k\omega_{\mathrm{can}}$ is exact, being d of $\frac{s^k}{k!}\iota_Z\mathscr{L}_Z^{k-1}\omega_{\mathrm{can}}$ (an easy induction using $\mathscr{L}_Z d = d\mathscr{L}_Z$). Integrating termwise in s from 0 to 1, we obtain

$$\omega = \omega_{\mathrm{can}} + \sum_{k=1}^{\infty} \lambda^k\, d\theta_k, \qquad \theta_k := -\int_0^1 \frac{(s\mathscr{L}_Z)^k}{k!}\, \theta_{\mathrm{can}}\, ds,$$

which is precisely the required expression (using $\omega_{\mathrm{can}} = -d\theta_{\mathrm{can}}$ to fix signs). This completes the proof. □

Moreover, the following theorem defines the classifying action:

Theorem 9. *The classifying action γ is given by $\gamma([B], [\pi]) = [\tau_{-B(\pi)}]$.*

Proof. Consider B and π representatives of $[B] \in H^2_{dR}(P)[[\lambda]]$ and $[\pi] \in \mathrm{FPois}_o(P)$. Let $\tau_{-B(\pi)}$ be a Poisson structure and, by Lemma III..3, consider the self-equivalence bimodule

$$(C^\infty(P)[[\lambda]], \pi) \xrightarrow{\Phi} (C^\infty(T^*P)[[\lambda]], \omega + \rho^*B) \xleftarrow{\Phi'} (C^\infty(P)[[\lambda]], \tau_{-B(\pi)}). \tag{3.44}$$

Again, by Lemma III..3, ω is cohomologous to ω_{can}, so $\omega + \rho^*B$ is cohomologous to $\omega_{\mathrm{can}} + \rho^*B$ and there is a Poisson isomorphism $\exp(\mathscr{L}_Z) : (C^\infty(T^*P)[[\lambda]], \omega + \rho^*B) \to (C^\infty(T^*P)[[\lambda]], \omega_{\mathrm{can}} + \rho^*B)$. As a result, we then obtain the following equivalence bimodule

$$(C^\infty(P)[[\lambda]], \pi) \xrightarrow{\exp(\mathscr{L}_Z)\rho^*} (C^\infty(T^*P)[[\lambda]], \omega_{\mathrm{can}} + \rho^*B) \xleftarrow{\mathscr{L}_Z\Phi'} (C^\infty(P)[[\lambda]], \tau_{-B(\pi)}), \tag{3.45}$$

which, by definition of γ, implies that $\tau_{-B(\pi)}$ is a representative of $\gamma([B], [\gamma])$. □

IV.. Main Result: Morita Equivalence vs. Gauge Equivalence for Formal Poisson Structures with $\pi_0 = 0$

1.. Statement of the theorem

Theorem 10. *Two Formal Poisson structures π and π' on P, vanishing at zeroth order, are Morita equivalent if and only if there exists a diffeomorphism $\psi \in \mathrm{Diff}(P)$ such that π and $\psi_*\pi'$ lie in the same orbit of the action of $H^2_{dR}(P)[[\lambda]]$ on $\mathrm{FPois}_0(P)$ by gauge transformations.*

Sketch of proof for Theorem 10. Assume that the formal Poisson structures π and π' on P are Morita equivalent. By Theorem 8, this Morita equivalence holds if and only if their classes in $FPois_0(P)$ are related by the equation $[\pi'] = \psi_*\gamma([B], [\pi])$ for some diffeomorphism $\psi \in Diff(P)$ and some formal de Rham cohomology class $[B] \in H^2_{dR}(P)[[\lambda]]$.

Next, Theorem 9 explicitly identifies the nature of the classifying action γ. It establishes that the action of $[B]$ on the class $[\pi]$ corresponds exactly to the gauge transformation given by $\gamma([B], [\pi]) = [\tau_{-B}(\pi)]$. Substituting the explicit formula from Theorem 9 into the equivalence established by Theorem 8, we immediately obtain $[\pi'] = \psi_*[\tau_{-B}(\pi)]$. This demonstrates that the two formal Poisson structures are Morita equivalent if and only if they differ by the combined action of a gauge transformation and a diffeomorphism, concluding the proof. □

Thus, one can think of Theorem 10 is as a direct corollary of the conjunction of the *classification* statement of (i) Proposition 8, which reduces Morita equivalence to a group action, and the *identification* statement of (ii) Proposition 9, which computes that action explicitly as B-field gauge transformation. The nontrivial mathematical content resides almost entirely in establishing (i) and (ii) separately: (i) is a deformation-theoretic argument through the vanishing of Chevalley–Eilenberg cohomology associated to $\rho : T^*P \to P$, while (ii) is a Courant-algebroid computation showing that the canonical self-bimodule of $(P, 0)$ deforms compatibly with B-fields.

2.. Conclusion and Future Directions

Theorem 13.1 shows that, once one restricts to formal Poisson structures deforming the trivial structure $\pi_0 = 0$, the geometric notion of Morita equivalence collapses onto the much more computable and classical operation of gauge transformation by closed 2-forms, modulo diffeomorphism. This is a very interesting and useful simplification relative to the classical (non-formal) theory, where Morita equivalence and B-field gauge equivalence of Poisson structures are known to diverge in general (cf. §5, Sec. 5 of [**?**]). The formal deformation parameter λ, therefore, forces every equivalence bimodule of $(P, 0)$-type formal Poisson structures to be modeled, order by order, on the cotangent bundle T^*P, and the associated Chevalley–Eilenberg cohomology controlling obstructions vanishes identically (Proposition 19). This unobstructedness is what ultimately collapses Morita theory into the action of $\mathrm{Diff}(P) \ltimes H^2_{\mathrm{dR}}(P)[[\lambda]]$.

The paper's closing observation — that Kontsevich's quantization map $\mathscr{K} : \mathrm{FPois0}(P) \xrightarrow{\sim} \mathrm{Def}(P)$ intertwines the gauge action of $H^2\mathrm{dR}(P)[[\lambda]]$ (equivalently, by Theorem [**?**], the Morita-equivalence orbit structure) with the action of $\mathrm{Diff}(P) \ltimes H^2(P, \mathbb{Z})$ on star-product Morita equivalence classes — suggests several directions in which Theorem [**?**] could be extended.

1. **Integrality and quantization of Morita classes.** The correspondence in Theorem 5.11 of [8] shows that Morita equivalence of star products corresponds, under $\mathscr{K}$, only to those B-fields lying in the image of $2\pi i H^2(P, \mathbb{Z}) \to H^2_{\mathrm{dR}}(P, \mathbb{C})$. A natural refinement of Theorem 10 would characterize precisely which formal-geometric Morita equivalences (in the sense of this paper) are "quantizable," i.e. arise from genuine Morita equivalences of associative star-product algebras, versus those that are only formally/geometrically valid. This would amount to an integral refinement of the classifying action γ, likely requiring a version of the theory over $H^2(P, \mathbb{Z})$-twisted (rather than merely $H^2_{\mathrm{dR}}(P)$-twisted) Poisson structures, connecting to gerbes and B-field quantization in the sense of Kapustin–Orlov-type noncommutative torus dualities.

2. **Beyond $\pi_0 = 0$: nontrivial zeroth-order Poisson structures.** The present results are confined to deformations of the zero Poisson structure, where the classical self-equivalence bimodules of $(P, 0)$ are classified completely [9]. For a general integrable Poisson manifold (P, π_0) with $\pi_0 \neq 0$, the analogous classical Morita self-equivalences are governed by the (typically nontrivial and non-abelian) Picard group $\mathrm{Pic}(P, \pi_0)$, and the Chevalley–Eilenberg cohomology $H^\bullet_{\mathrm{CE,der}}(P, S)$ controlling deformation of Poisson morphisms need not vanish. Extending Theorem 10 to this setting would require replacing the abelian group $H^2_{\mathrm{dR}}(P)[[\lambda]]$ by a suitable formal deformation of $\mathrm{Pic}(P, \pi_0)$, and would connect Morita theory of formal Poisson structures to the deformation theory of symplectic groupoids integrating π_0.

3. **Morita equivalence of A_∞/curved deformations and homological mirror symmetry.** Since Hochschild cohomology $HH^\bullet(C^\infty(P))$ governs deformation quantization via the Hochschild–Kostant–Rosenberg (HKR) isomorphism $HH^\bullet(C^\infty(P)) \cong \mathfrak{X}^\bullet(P)$, and Morita equivalence of associative algebras is classically characterized via Hochschild cohomology and Morita-invariant Ext/Tor data, a natural project is to reformulate the classifying action

γ directly at the level of Hochschild cochains, i.e., to realize the $H^2_{\mathrm{dR}}(P)[[\lambda]]$-action of Theorem 10 as the shadow, under HKR, of an action on the Maurer–Cartan/Deligne groupoid controlling A_∞-deformations of $C^\infty(P)[[\lambda]]$ as a curved A_∞-algebra. This would place Theorem 10 inside the broader homotopical framework of derived Morita theory and could interact productively with the noncommutative-geometric perspective on Poisson bivectors as "first-order noncommutative directions" central to homological mirror symmetry (Kontsevich's original motivation for formality).

4. **Groupoid and stacky refinements.** Since Morita equivalence of Poisson manifolds is tightly linked to Morita equivalence of the symplectic groupoids integrating them, and star-product Morita equivalence is linked to Morita equivalence of the corresponding deformation-quantization algebroids/gerbes, a further avenue is to lift Theorem 10 to statement about formal deformations of symplectic groupoids themselves (rather than just the underlying Poisson manifolds), potentially yielding a version of the classifying action γ valid at the level of differentiable stacks, connecting to Cattaneo–Felder-type constructions of the symplectic groupoid via the Poisson sigma model.

Taken together, these directions suggest that Theorem 10, while formulated purely in the formal-geometric category, is best understood as the "classical shadow" of a richer story linking Hochschild-cohomological deformation theory, noncommutative Morita theory, and the arithmetic/integral structure of B-field gauge transformations — a story whose noncommutative-algebraic completion remains open beyond the star-product correspondence already established via Kontsevich's map $\mathscr{K}$.

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